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FinSight

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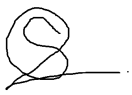
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Abstract

Forecasting the KSE-100 remains a challenge for investors in Pakistan, as existing models rely mostly on past prices, ignore the news that moves the market, and treat bull and bear markets identically. This project introduces FinSight, a system that combines natural language processing and deep learning to forecast the index. A DistilBERT model converts Dawn Business headlines into a daily sentiment score, while K-Means clustering labels each trading day as bull, bear, or sideways. These signals, along with the Pakistan EPU index and technical indicators, feed an LSTM that produces 3-day, 7-day, and 14-day forecasts, explained per regime using SHAP. FinSight gives investors transparent, data-driven forecasts and is an important step toward smarter decision-making in Pakistan's stock market.

Executive Summary

Investors in Pakistan's stock market often make decisions on the KSE-100 without any AI-assisted tooling. Existing forecasting models rely mostly on past prices and ignore the news that moves the market. They also treat bull and bear markets identically, using the same model and features whatever the market conditions, and, to our knowledge, no labeled Pakistani financial news dataset exists to support better models. To address this, we propose FinSight, an end-to-end system that forecasts the KSE-100 index while accounting for news sentiment, market regime, and macroeconomic uncertainty.

FinSight collects business news from Dawn Business and converts it into a daily sentiment score. A DistilBERT model, fine-tuned and evaluated on 24,583 labeled headlines, reaches 82.7% test accuracy and classifies each headline as positive, negative, or neutral. The fine-tuned model is then used to label the full collected corpus of 37,806 headlines. Pretrained FinBERT was found to reverse the direction of metrics such as deficit, inflation, and circular debt, which is why a domain-specific labeled dataset was built. In parallel, K-Means clustering on rolling return and volatility labels every trading day as bull, bear, or sideways, so the forecasting model knows what kind of market it is predicting.

These signals are combined with the Pakistan Economic Policy Uncertainty index and technical indicators (RSI, MACD, and EMA) in a unified dataset. A hybrid LSTM network reads the last 30 days of this data to produce 3-day, 7-day, and 14-day forecasts, with uncertainty estimated through Monte Carlo dropout. To avoid a black-box output, SHAP is applied separately for each market regime, showing which signals drive a forecast in bull, bear, and sideways conditions. The results will be delivered through a web application with a live dashboard, multi-horizon forecasts, a sentiment feed, an explainability panel, and a REST API for integration with other fintech applications. The current phase focuses on the backend pipeline, and the web application is planned for the next phase.

FinSight aims to make AI-assisted forecasting accessible to retail investors, portfolio managers, and fintech practitioners alike. Investors get forecasts they can understand, with the reasons behind each one, instead of an unexplained number. Researchers gain what we believe is the first labeled Pakistani financial news sentiment dataset and a regime-aware forecasting framework that can be extended to other emerging markets.

In conclusion, FinSight represents a significant step toward intelligent, transparent forecasting for Pakistan's equity market. By combining natural language processing, regime detection, deep learning, and explainable AI in a single pipeline, it offers a solution that benefits both market participants and the research community.

Table of Contents

List of Figures **ix**

List of Tables **x**

1 Introduction **1**

1.1 Purpose of this Document 1

1.2 Intended Audience 2

1.3 Definitions, Acronyms, and Abbreviations 2

1.4 Conclusion 3

2 Project Vision **4**

2.1 Problem Domain Overview 4

2.2 Problem Statement 4

2.3 Problem Elaboration 4

2.4 Goals and Objectives 5

2.4.1 Goal 5

2.4.2 Objectives 5

2.5 Project Scope 6

2.6 Sustainable Development Goal (SDG) 7

2.7 Constraints 7

2.8 Business Opportunity 8

2.9 Stakeholders Description/ User Characteristics 8

2.9.1 Stakeholders Summary 8

2.9.2 Key High-Level Goals and Problems of Stakeholders 8

3 Literature Review / Related Work **10**

3.1 Definitions, Acronyms, and Abbreviations 10

3.2 Detailed Literature Review 11

3.2.1 Sentiment-Augmented and Hybrid LSTM Forecasting 11

3.2.2 Financial News Sentiment Modelling . . . 13
3.2.3 Explainable AI for Stock Prediction . . . 14
3.2.4 Market Regime Detection and Regime-Aware Forecasting . . . 15
3.2.5 Forecasting Studies on the Pakistani Market . . . 16
3.3 Literature Review Summary Table . . . 17
3.4 Related Products and Applications . . . 19
3.4.1 PSX Data Portal . . . 19
3.4.2 Sarmaaya.pk . . . 20
3.4.3 Capital Stake Analytics Platform . . . 20
3.4.4 Mettis Global . . . 21
3.4.5 UpInvest . . . 21
3.4.6 TradingView . . . 21
3.4.7 Comparison with FinSight . . . 22
3.5 Conclusion . . . 23

4 Software Requirement Specifications 24
4.1 List of Features . . . 24
4.2 Functional Requirements . . . 24
4.2.1 Functional Requirements for Visitor . . . 24
4.2.2 Functional Requirements for System . . . 25
4.3 Quality Attributes . . . 25
4.4 Non-Functional Requirements . . . 26
4.5 Assumptions . . . 26
4.6 Use Cases . . . 26
4.7 Hardware and Software Requirements . . . 28
4.7.1 Hardware Requirements . . . 28
4.7.2 Software Requirements . . . 28
4.8 Graphical User Interface . . . 29
4.9 Database Design . . . 31
4.9.1 ER Diagram . . . 32
4.9.2 Data Dictionary . . . 33
4.10 Risk Analysis . . . 35
4.11 Conclusion . . . 35

5 Proposed Approach and Methodology 36

5.1 Overview of the Approach . . . 36
5.2 Data Collection and Preprocessing . . . 38
5.2.1 KSE-100 Price Data . . . 38
5.2.2 News Headlines . . . 39
5.2.3 Macroeconomic Data . . . 39
5.3 Sentiment Analysis . . . 39
5.3.1 Labelled Dataset Construction . . . 39
5.3.2 Fine-Tuning DistilBERT . . . 40
5.3.3 Daily Sentiment Aggregation . . . 40
5.4 Technical Indicators . . . 41
5.5 Market Regime Detection . . . 41
5.6 Dataset Assembly . . . 42
5.7 Hybrid LSTM Forecasting Model . . . 42
5.7.1 Model Structure . . . 42
5.7.2 Multi-Horizon Forecasting . . . 43
5.7.3 Confidence Intervals . . . 43
5.8 Regime-Wise Explainability . . . 43
5.9 Evaluation Methodology . . . 44
5.10 Conclusion . . . 44

6 High-Level and Low-Level Design 45
6.1 System Overview . . . 45
6.2 Design Considerations . . . 45
6.2.1 Assumptions and Dependencies . . . 45
6.2.2 General Constraints . . . 46
6.2.3 Goals and Guidelines . . . 46
6.2.4 Development Methods . . . 47
6.3 System Architecture . . . 47
6.3.1 Subsystem Architecture . . . 49
6.4 Architectural Strategies . . . 50
6.4.1 Layered Architecture . . . 50
6.4.2 Offline Pipeline and Precomputed Results . . . 50
6.4.3 Relational Database as the Single Data Store . . . 50
6.4.4 REST API Between Backend and Frontend . . . 50
6.4.5 Uncertainty Through Monte Carlo Dropout . . . 51

6.4.6 Regime-Wise Explainability 51
6.4.7 Error Detection and Recovery 51
6.4.8 Future Extension 51
6.5 Domain Model/Class Diagram 51
6.5.1 Processing Classes 52
6.5.2 Data Classes and Relationships 53
6.6 Policies and Tactics 53
6.6.1 Choice of Technology 53
6.6.2 Data Handling Policy 54
6.6.3 Testing and Validation Policy 54
6.6.4 Maintenance Policy 54
6.7 Conclusion 54

List of Figures

2.1 **SDG 8 and SDG 9** 7

4.1 **Dashboard screen showing live KSE-100 prices, indicators and market regime** . . . 29
4.2 **Forecast screen showing 3, 7 and 14 day forecasts with confidence intervals** 30
4.3 **Sentiment feed screen showing headlines with sentiment labels** 30
4.4 **Explainability screen showing SHAP importance for each market regime** 31
4.5 **GUI navigation flow between the dashboard, forecast, sentiment feed and explainability screens** 31
4.6 **ER diagram. It describes the database design of our system.** 33

5.1 **Proposed workflow of FinSight, from data collection to forecasts, explanations, and storage.** 38

6.1 **System architecture of FinSight showing the data, ML pipeline, storage and application layers** 48
6.2 **Class diagram of FinSight showing the backend processing classes and their data classes** 52

List of Tables

3.1 Summary of the reviewed research studies. 17
3.2 Comparison of related products and applications with FinSight. 22

4.1 Use case: View KSE-100 Dashboard. 27
4.2 Use case: View Multi-Horizon Forecast. 27
4.3 Use case: View Daily Sentiment Feed. 27
4.4 Use case: View Explanation by Market Regime. 28
4.5 Data dictionary: news_article. 33
4.6 Data dictionary: article_sentiment. 33
4.7 Data dictionary: daily_sentiment. 34
4.8 Data dictionary: market_data. 34
4.9 Data dictionary: technical_indicator. 34
4.10 Data dictionary: market_regime. 34
4.11 Data dictionary: macro_epu. 34
4.12 Data dictionary: forecast. 34
4.13 Data dictionary: shap_value. 35

Chapter 1 Introduction

Investing in Pakistan's stock market is difficult for ordinary participants. The KSE-100, the benchmark index of the Pakistan Stock Exchange (PSX), is highly sensitive to political events, monetary policy, and economic news, and moves sharply on all of them. Retail investors, portfolio managers, and fintech firms make decisions on it with little to no AI-assisted tooling, even though hybrid forecasting systems that combine news sentiment, macroeconomic signals, and deep learning are an active area of research worldwide.

Two gaps keep such systems from reaching Pakistan. First, to our knowledge no labeled Pakistani financial news sentiment dataset exists, and general-purpose sentiment models such as FinBERT do not transfer reliably to local news. In our audit, FinBERT reversed the direction of metrics such as deficit, inflation, and circular debt at high confidence. Second, existing KSE-100 forecasting models ignore market regime, predicting bull and bear markets with the same model and features even though the market behaves very differently in each.

FinSight is our solution to both gaps. It is an end-to-end pipeline that forecasts the KSE-100 index at 3-day, 7-day, and 14-day horizons. A DistilBERT model fine-tuned on a labeled corpus of Dawn Business headlines turns daily news into a sentiment score. K-Means clustering on rolling return and volatility labels each trading day as bull, bear, or sideways. These signals, together with the Pakistan Economic Policy Uncertainty (EPU) index and technical indicators, feed a hybrid LSTM forecasting model. SHAP is then applied separately for each regime to show which signals drive a forecast under each market condition.

1.1 Purpose of this Document

This document presents the problem, background, design, implementation, and evaluation of FinSight. It covers the construction of the Pakistani financial sentiment dataset, the regime detection method, the hybrid LSTM forecasting model, and the regime-conditioned explainability analysis.

The main research questions we aim to answer are: Does adding news sentiment, market regime, and macroeconomic uncertainty improve KSE-100 forecasts over models that use price data alone? Do the features that drive forecasts differ between bull, bear, and sideways markets? This report presents the methodology, design, implementation, testing, and evaluation of the project, as well as its limitations and future work.

1.2 Intended Audience

This document is intended for the FYP panel and our supervisor, who will analyze and evaluate our approach. Researchers in financial NLP and time-series forecasting can use it as a starting point for work on emerging markets, since it describes the dataset construction, the labeling decisions, and the modeling choices in detail. Finally, investors and fintech practitioners in Pakistan who are interested in AI-assisted decision tools will find an overview of what the system can and cannot do.

1.3 Definitions, Acronyms, and Abbreviations

In the interest of clarity and readability, this report uses several acronyms. Their full forms are listed below.

API: Application Programming Interface

BERT: Bidirectional Encoder Representations from Transformers

DistilBERT: Distilled version of BERT

EMA: Exponential Moving Average

EPU: Economic Policy Uncertainty

ER: Entity-Relationship

FYP: Final Year Project

GUI: Graphical User Interface

KSE-100: Karachi Stock Exchange 100 Index

LSTM: Long Short-Term Memory

MACD: Moving Average Convergence Divergence

MAE: Mean Absolute Error

MAPE: Mean Absolute Percentage Error

NLP: Natural Language Processing

OHLCV: Open, High, Low, Close, Volume

PSX: Pakistan Stock Exchange

REST: Representational State Transfer

RMSE: Root Mean Squared Error

RSI: Relative Strength Index

SDG: Sustainable Development Goal

SHAP: SHapley Additive exPlanations

1.4 Conclusion

This document is structured as follows. Chapter 1 introduces the project, its purpose, its audience, and the terms used. Chapter 2 describes the project vision, including the problem statement, goals and objectives, scope, and constraints. Chapter 3 reviews related work on financial sentiment analysis, regime detection, LSTM forecasting, and explainable AI. Chapter 4 specifies the software requirements, including features, use cases, and risks. Chapter 5 describes the proposed approach and methodology. Chapter 6 presents the system design and architecture. Chapter 7 covers the implementation and test cases. Chapter 8 is the user manual. Chapter 9 presents the experimental results and discussion, and Chapter 10 concludes the report.

Chapter 2 Project Vision

Pakistani investors, portfolio managers, and fintech firms need forecasting tools that reflect local news, local macroeconomic conditions, and the market state the KSE-100 is currently in. FinSight is a regime-aware, sentiment-augmented forecasting system built to provide them. This chapter defines the problem, the goals and scope of the project, the Sustainable Development Goals it supports, and the stakeholders it serves.

2.1 Problem Domain Overview

FinSight is an end-to-end system that forecasts the KSE-100 index at 3-day, 7-day, and 14-day horizons and explains each forecast. It scrapes business news from Dawn Business and converts it into a daily sentiment score using a DistilBERT model fine-tuned on a labeled Pakistani financial news dataset built for this project. It labels every trading day as bull, bear, or sideways by clustering 20-day rolling return and volatility with K-Means. These signals are combined with the Pakistan Economic Policy Uncertainty (EPU) index and technical indicators (RSI, MACD, and EMA) in a single dataset. A hybrid LSTM network reads a 30-day window of this dataset to produce the forecasts, and SHAP is applied separately for each market regime to show which signals drive a prediction under each condition. The results are planned to be delivered through a web application with a dashboard, multi-horizon forecasts, a sentiment feed, an explainability panel, and a REST API, to be developed after the back-end pipeline is complete.

2.2 Problem Statement

Pakistan's equity market has no AI-assisted forecasting tool that combines local news sentiment, macroeconomic uncertainty, and market regime context, and its forecasts cannot be explained to the investors who rely on them.

2.3 Problem Elaboration

The problem has several sub-problems, each of which the project addresses.

- No labeled Pakistani sentiment dataset. Existing financial sentiment resources use English or Indian news and do not reflect Pakistan's market language. General-purpose models such as FinBERT also make systematic errors on local news, for example reversing the direction of deficit, inflation, and circular debt headlines at high confidence. A new labeled dataset and a fine-tuned

model are required.

- Regime-blind forecasting. Existing KSE-100 models predict bull and bear markets with the same model and features, although the market behaves very differently in each. The market state must be detected and given to the forecasting model as an input.
- Fragmented signals. Prior Pakistani studies use price indicators, sentiment, or EPU separately, but to our knowledge none combines them in one deep learning pipeline.
- Opaque predictions. Deep learning forecasts come without reasons, which makes them hard to trust in financial decisions. Explanations that apply across all market conditions can hide the fact that the drivers change between regimes.
- No accessible tooling. Retail investors and small fintech firms in Pakistan have no public interface or API for AI-assisted KSE-100 forecasts.

2.4 Goals and Objectives

2.4.1 Goal

The goal of this project is to build and evaluate an explainable, regime-aware forecasting system for the KSE-100 that uses local news sentiment and macroeconomic uncertainty, and to make it available to investors and fintech practitioners in Pakistan.

2.4.2 Objectives

- To build a labeled Pakistani financial news sentiment dataset from Dawn Business headlines and fine-tune DistilBERT on it.
- To detect bull, bear, and sideways market regimes from KSE-100 prices using K-Means clustering.
- To train a hybrid LSTM model that forecasts the KSE-100 at 3-day, 7-day, and 14-day horizons with Monte Carlo dropout uncertainty.
- To evaluate whether sentiment, regime, and EPU features improve forecasts over a price-only baseline.
- To apply SHAP separately for each regime to identify which features drive forecasts in each market condition.

2.5 Project Scope

The scope of this project is to design, develop, and evaluate a system that forecasts the KSE-100 index using news sentiment, market regime, macroeconomic uncertainty, and technical indicators. The system will:

- Collect and process Dawn Business news and KSE-100 daily price data
- Classify headline sentiment as positive, negative, or neutral and aggregate it into a daily score
- Assign each trading day to a bull, bear, or sideways regime
- Produce 3-day, 7-day, and 14-day KSE-100 forecasts with uncertainty estimates
- Explain forecasts using SHAP, separately for each regime
- Provide the forecasting back-end, with a web dashboard and REST API designed as a planned extension

The project is developed in Python, using HuggingFace Transformers for sentiment modeling, scikit-learn for clustering, and a deep learning framework for the LSTM.

The project deliverables include:

- The labeled Pakistani financial news sentiment dataset and the fine-tuned DistilBERT model
- The regime detection module and the trained forecasting model
- The SHAP analysis per market regime
- A documented back-end pipeline exposing the forecasts and explanations (web application and REST API as future work)
- This report documenting the design, implementation, and evaluation

The project scope does not include:

- Forecasting individual stocks, as the system targets the KSE-100 index only
- Providing investment advice or automated trading
- Developing native mobile applications (iOS or Android)
- Providing real-time intraday forecasts, as the system works on daily data
- Full development and public deployment of the web application within this project phase

This scope is used as a guide throughout development.

2.6 Sustainable Development Goal (SDG)

This project targets SDG 9 (Industry, Innovation, and Infrastructure) and SDG 8 (Decent Work and Economic Growth). It supports SDG 9 by applying artificial intelligence to Pakistan's financial infrastructure and by providing open datasets and methods that other researchers can extend. It supports SDG 8 by giving retail investors and small fintech firms access to forecasting tools that were previously available only to large institutions, which encourages informed participation in the capital market. The complete SDG list is shown in Figure 2.1.



Figure 2.1: SDG 8 and SDG 9

2.7 Constraints

The project operates under the following constraints.

- Data availability. Pakistani financial news is not available through an API, so Dawn Business articles had to be scraped. The Wayback Machine is blocked on Pakistani ISPs, so archived pages could not be used as a fallback.
- Data quality. The KSE-100 volume series had about 40% missing values, concentrated between 2019 and 2024, so volume was dropped from the feature set. The EPU index is monthly, so each monthly value is carried over to all days of its month.
- Annotation effort. Sentiment labels were created through FinBERT-assisted manual annotation by annotators without professional financial training.
- Compute resources. Model training is limited to the GPU and storage available through Google Colab and Google Drive.

2.8 Business Opportunity

Pakistan's capital market is growing, yet retail investors and small firms have few analytical tools built for local conditions. International platforms provide generic charts and use models trained on other markets and on English news. FinSight can be offered as a subscription dashboard for retail investors, as an API for brokerage houses and fintech applications, and as a data service built on its sentiment feed and regime labels. The labeled sentiment dataset and model also have value for researchers and financial media who need Pakistani financial text analysis. The regime-aware, explainable design also differentiates the product from black-box forecasting tools.

2.9 Stakeholders Description/ User Characteristics

FinSight has several groups of users, each with different technical backgrounds and needs.

2.9.1 Stakeholders Summary

- Retail investors: individuals trading on the PSX who need plain-language forecasts and a simple interface. They typically have limited technical background.
- Portfolio managers and analysts: professionals who need multi-horizon forecasts, uncertainty bands, and explanations they can justify to clients.
- Fintech developers: teams that want to integrate forecasts and sentiment scores into their own applications through the REST API.
- Researchers and students: users of the dataset, the sentiment model, and the regime-conditioned explainability analysis.
- FYP supervisor and evaluation panel: reviewers who assess the technical quality of the work.

2.9.2 Key High-Level Goals and Problems of Stakeholders

- Retail investors want to know where the market may be heading and why, in terms they understand. Their problem is that existing tools give unexplained predictions or none at all.
- Portfolio managers want forecasts they can trust and defend. Their problem is that black-box models give no reasons, and models trained on other markets do not reflect Pakistani conditions.
- Fintech developers want simple, reliable endpoints for forecasts and sentiment. Their problem is that, to our knowledge, no Pakistani source for such data exists.

- Researchers want open, labeled Pakistani financial text and reproducible methods. Their problem is that no such dataset has, to our knowledge, been published.
- The FYP panel wants a technically sound, novel, and well-documented project. Their concern is that the claims are supported by evidence.

Chapter 3 Literature Review / Related Work

This chapter reviews sixteen studies and six existing products and applications relevant to FinSight, a regime-aware, sentiment-augmented and explainable forecasting system for the KSE-100 index. The studies are grouped into five categories: (i) sentiment-augmented and hybrid LSTM forecasting, (ii) financial news sentiment modelling, (iii) explainable AI for stock prediction, (iv) market regime detection and regime-aware forecasting, and (v) existing work on the Pakistani market. For each study we give a summary, a critical analysis of its strengths and weaknesses, and its relationship to the proposed work. A summary table follows the detailed review. The chapter then discusses the related products and applications available to investors in Pakistan, describes their features, and compares them with FinSight. A concluding section ends the chapter.

3.1 Definitions, Acronyms, and Abbreviations

ARIMA AutoRegressive Integrated Moving Average, a classical linear time-series forecasting model.

BiLSTM Bidirectional Long Short-Term Memory network.

DBSCAN Density-Based Spatial Clustering of Applications with Noise.

DistilBERT A distilled, lighter version of the BERT transformer language model.

EMA Exponential Moving Average, a trend-following technical indicator.

EMD Empirical Mode Decomposition, a signal decomposition technique.

EPU Economic Policy Uncertainty index.

FinBERT A BERT model pre-trained and fine-tuned on financial text.

FIFO First In, First Out, a method of matching sold shares against the earliest purchased shares when calculating profit and loss.

GARCH / GARCH-MIDAS Generalized Autoregressive Conditional Heteroskedasticity models; the MIDAS variant mixes data sampled at different frequencies.

GMM Gaussian Mixture Model.

HMM Hidden Markov Model.

KF / MS Kalman Filter / Markov Switching model.

KSE-100 Karachi Stock Exchange 100 Index, the benchmark index of the Pakistan Stock Exchange.

LIME Local Interpretable Model-agnostic Explanations.

LSTM Long Short-Term Memory network.

OHLCV Open, High, Low, Close, Volume price data.

PCA Principal Component Analysis.

PSX Pakistan Stock Exchange.

RSI Relative Strength Index, a momentum indicator.

SHAP SHapley Additive exPlanations.

XAI Explainable Artificial Intelligence.

3.2 Detailed Literature Review

3.2.1 Sentiment-Augmented and Hybrid LSTM Forecasting

A large body of work combines news sentiment with recurrent networks to forecast equity prices. The five studies below cover transformer-based sentiment with LSTM, decomposition-based hybrids, and technical-indicator LSTMs. Together they establish the forecasting backbone that FinSight builds on.

3.2.1.1 FinBERT–LSTM for Stock Movement Prediction

Jiang and Zeng [1] used FinBERT to extract sentiment from stock-market news and fed it into an LSTM to forecast market movement. They compared the resulting model against standard BERT, a standalone LSTM and the traditional ARIMA model. Their findings indicate that adding sentiment analysis significantly improves the model's ability to anticipate market fluctuations, and they propose several optimisation techniques to refine performance further.

The strength of this study is its clear comparison against both classical and neural baselines, which isolates the contribution of sentiment. Its weaknesses are that it relies on English-language news from a general financial corpus and treats all market conditions identically, with no regime information and no explanation of which inputs drive a forecast.

This work is the closest general template for FinSight's pipeline of sentiment scoring followed by LSTM forecasting. FinSight differs by replacing FinBERT with DistilBERT fine-tuned on Pakistani headlines, conditioning on market regime, and adding per-regime explainability.

3.2.1.2 FinBERT and LSTM for S&P 500 Forecasting

Kim et al. [2] applied FinBERT to summarised New York Times news text and combined the resulting sentiment scores with S&P 500 index data in an LSTM. The sentiment scores were aligned to business-

day index data, and the authors show that using summarised text rather than entire articles is an effective strategy for sentiment-based forecasting.

A strength is the demonstration that short, condensed text is sufficient, which lowers data-collection and compute cost. The study is limited to a single US index and a single US news source, and offers neither regime awareness nor explainability.

This finding supports FinSight's decision to use short text (a headline and its lead paragraph) scraped from one source rather than full articles, and its daily alignment of sentiment scores to trading days.

3.2.1.3 PCA–EMD–LSTM with Sentiment on the Thai Stock Market

Srijiranon et al. [3] proposed PCA-EMD-LSTM, a hybrid that combines principal component analysis, empirical mode decomposition and LSTM to predict the next-step closing price of the Thai stock market. They added FinBERT-based sentiment from financial and economic news and tested on 2018–2022 data. The proposed framework performed best against the baselines, and sentiment improved the original LSTM.

The study shows that hybrid pre-processing and news sentiment help in a non-US, emerging-market setting. However, it does not appear to condition on market regimes or provide explanations for its predictions, and its decomposition-heavy pipeline is harder to interpret than a plain LSTM.

For FinSight, this is evidence that sentiment-augmented LSTMs transfer to emerging markets, while the lack of regime handling and interpretability reinforces the gaps that FinSight targets.

3.2.1.4 Hybrid RNN–LSTM with Sentiment on the BSE Sensex

Kasture and Shirsath [4] integrated text-derived sentiment scores with historical BSE Sensex data in a hybrid RNN-LSTM model. They compared it against Random Forest and Support Vector regressors, and against the same LSTM trained without sentiment scores. Their results show that incorporating sentiment enriches the predictive capability of the model.

The ablation against a sentiment-free LSTM and classical regressors is a strength. The study covers only one Indian index and neither accounts for regime shifts nor explains predictions.

It provides further South Asian evidence that sentiment features improve LSTM forecasts, which FinSight extends with a Pakistan-specific sentiment dataset.

3.2.1.5 LSTM with EMA and RSI Technical Indicators

Dhokane and Agarwal [5] combined the RSI with 50-day, 100-day and 150-day EMAs and basic OHLCV features as inputs to an LSTM predicting closing prices, using NSE India data obtained through Yahoo Finance. They evaluated the model with RMSE, MAPE and R^2 and report improved fit with the indicators included.

The approach is simple and reproducible, but it uses price data only, with no textual or macroeconomic signal, no regime information and no explainability.

FinSight adopts a similar technical-indicator feature block (including EMA and RSI) and a similar public daily OHLCV source, and extends it with sentiment, regime and uncertainty features.

3.2.2 Financial News Sentiment Modelling

Reliable sentiment scores are the prerequisite for sentiment-augmented forecasting. The two studies below address transformer-based classification of financial news and the link between aggregate news sentiment and market returns in South Asia.

3.2.2.1 DistilBERT for Banking Financial News

Dogra et al. [6] compared the contextual language model DistilBERT with the traditional context-independent TF-IDF representation for three-class (positive, negative, neutral) sentiment classification of Indian banking, governmental and global news events.

A strength is that it evaluates DistilBERT, a lightweight transformer, on South Asian financial news. It is, however, a classification study focused on the banking domain; it is not linked to index-level price forecasting.

It provides direct precedent for FinSight's choice of DistilBERT as a practical, efficient model to fine-tune for headline sentiment.

3.2.2.2 Aggregate News Sentiment and Stock Returns in India

Chari et al. [7] examined the relationship between aggregate (market-wide) news sentiment and stock market returns on days of significant market movement, using GARCH modelling, regression and dictionary-based sentiment analysis. They report that aggregate news sentiment has a short-lived influence, with notable effects from the business and politics categories.

The focus on aggregate rather than firm-specific sentiment is a strength and matches an index-level forecasting task. Its dictionary-based sentiment method cannot capture context, and the analysis is

restricted to days of significant market movement.

The short-lived effect supports FinSight's daily sentiment feed and short forecasting horizons, while the limits of dictionary methods motivate a fine-tuned transformer.

3.2.3 Explainable AI for Stock Prediction

Explainability techniques such as SHAP and LIME have been applied to stock predictors to expose which inputs drive a forecast. The three studies below represent this line of work.

3.2.3.1 Explainable Deep Learning for Stock Trend Prediction

Muhammad et al. [8] proposed a deep learning model that predicts five stock trend classes (upward, downward, double top, rounded bottom and rounded top) across four real-world datasets. They report an average accuracy of 94.9% against 85.7% for random forest, 60.07% for SVM and 52.45% for logistic regression, and use SHAP and LIME to interpret the model. SHAP identified the ten most influential features, enabling feature reduction while performance was maintained.

The combination of strong benchmarks and interpretability is a strength. The task is trend classification rather than price forecasting, and the explanations are computed globally across all predictions without conditioning on market state.

FinSight borrows the use of SHAP for feature importance and feature reduction, but computes it separately for each market regime.

3.2.3.2 Explainable AI with Deep Neural Networks for Stock Prediction

Wallapure Manikrao et al. [9] combined deep neural network forecasting (including LSTM) with XAI methods, namely SHAP and LIME, to make stock market predictions interpretable.

The work demonstrates that deep sequence models can be paired with post-hoc explanations without abandoning predictive power. As with the previous study, the explanations are, to the best of our reading, global rather than regime-specific, and sentiment from local news is not a central input.

It supports the feasibility of explainable LSTM forecasting in FinSight and highlights the missing regime-conditioned explanation layer.

3.2.3.3 FinBERT Sentiment with SHAP and Differential Privacy

Ruan and Jiang [10] combined FinBERT sentiment extracted from news headlines with technical and statistical indicators in an XGBoost classifier to forecast short-term movement of S&P 500 equities over

2018–2023. Their model outperformed technical-only and lexicon-based sentiment baselines on AUC, F1-score and simulated trading profitability, and SHAP analysis confirmed that FinBERT-derived features rank among the most influential predictors. The framework also incorporates differential privacy.

A strength is the direct SHAP evidence that transformer sentiment is a useful predictor. Its limitations are the use of a gradient-boosted tree instead of a sequence model, a US-only sample, and global rather than regime-conditioned explanations.

This study validates FinSight's plan to show how much headline sentiment contributes to each forecast, and strengthens the case for transformer sentiment over lexicon sentiment.

3.2.4 Market Regime Detection and Regime-Aware Forecasting

The central research question of FinSight is whether regime information improves forecasting. The three studies below provide the methodological foundation for detecting and using regimes.

3.2.4.1 Regime-Aware LightGBM with Walk-Forward Validation

Pagliaro [11] proposed a LightGBM framework conditioned on bull, sideways and bear regimes detected with a rolling Hidden Markov Model. Validation uses a walk-forward protocol with 100 expanding-window folds and 10-day purge windows to prevent leakage. Across a 51-stock universe the out-of-fold accuracy averaged 54.2%, Sharpe ratios were positive but not significant after Deflated Sharpe Ratio correction, and SHAP revealed regime-dependent decision logic.

The rigorous leakage control and the honest reporting of statistically modest results are notable strengths. The sample is US equities with cross-asset features, the model is a tree ensemble rather than a neural sequence model, and no sentiment or emerging-market data are used.

This is the closest methodological ancestor of FinSight. FinSight adopts regime-specific SHAP and walk-forward validation ideas for the forecasting stage, but uses unsupervised clustering instead of an HMM, an LSTM instead of LightGBM, and adds Pakistani sentiment and a macroeconomic uncertainty feature.

3.2.4.2 Kalman Filter, Markov Switching and LSTM Across Three Indices

Tian et al. [12] proposed a hybrid of Kalman filtering, Markov Switching and LSTM/GRU networks (KF–MS–LSTM), evaluated on the CSI 300, S&P 500 and Nikkei 225 indices. Regime probabilities from a three-state Markov Switching model were fed to the recurrent networks. The authors conclude that regime conditioning is the primary source of forecasting and economic improvement, and that models without regime information, including pure LSTM/GRU and a standalone Transformer, generally

performed worse.

Cross-market validation and the explicit use of forward-filtered probabilities, which avoid look-ahead bias, are strengths. The study uses no text sentiment, offers no explainability, and does not include a frontier market such as Pakistan.

This is the strongest direct empirical support for FinSight's hypothesis that regime inputs improve LSTM forecasts. It also warns that regime labels should be computed using only information available at each date. FinSight's regime descriptors use past prices only, and the remaining look-ahead in the cluster centroids is discussed in Chapter 5.

3.2.4.3 Clustering-Based Regime Identification

Ampountolas [13] applied K-Means, DBSCAN and Gaussian Mixture Models to daily data from July 2022 to February 2025 to identify distinct market regimes across traditional and digital assets during the 2024 U.S. election cycle, showing how political events shape market states.

Unsupervised clustering needs no restrictive parametric assumptions and yields interpretable regimes, which is a strength. The study is descriptive rather than predictive, covers a short window, and focuses on US and cryptocurrency markets.

It supports FinSight's use of unsupervised clustering to label bull, bear and sideways regimes directly from price behaviour.

3.2.5 Forecasting Studies on the Pakistani Market

Work on Pakistan is limited and, in the studies below, does not combine sentiment, regime awareness and explainability.

3.2.5.1 Economic Policy Uncertainty and Pakistani Market Volatility

Ghani and Ghani [14] studied how economic policy uncertainty indices for Pakistan, the US, China and the UK affect Pakistan's stock market volatility, using GARCH-MIDAS models, combination forecasts and Model Confidence Set tests. They found the US EPU index to be the most powerful predictor, the UK index to be informative, and the Pakistan and China indices to have no significant predictive information, with consistent results during COVID-19.

The study is rigorous on volatility prediction and robustness. It forecasts volatility rather than price level or direction and uses no deep learning or sentiment.

It motivates FinSight's macroeconomic uncertainty feature, and its finding about domestic versus US

uncertainty is a caution to be tested when evaluating that feature.

3.2.5.2 Sentiment-Based LSTM for Pakistan Stock Exchange Banks

Mehmood and Ali [15] predicted the next 30 days of closing prices for two banks listed on the PSX using LSTM and a BiLSTM with self-attention. News sentiment, computed with the Python NLTK module, was combined with ten years of historical data (January 2011 to July 2021), and adding sentiment reduced the mean squared error.

This is the most similar Pakistani precedent for sentiment-LSTM forecasting. Its sentiment is lexicon-based, it covers only two stocks, and it has no regime awareness or explainability.

FinSight extends this idea to the KSE-100 index with a transformer fine-tuned on a Pakistani headline dataset, regime conditioning and per-regime explanations.

3.2.5.3 Machine Learning and Technical Indicators on the PSX

Raza and Akhtar [16] used machine learning models (ANN, SVM and Random Forest) with 27 technical indicators to predict stock prices in the Pakistan Stock Exchange, finding that indicators such as %R, Momentum and Disparity 5 are important across all models. They explicitly suggest hybrid models with real-time data, sentiment analysis and external factors as future work.

The broad indicator set and cross-model importance analysis are strengths. The study uses technical data only, with no sentiment, regime or explainability component.

Its suggested future work matches the FinSight design, and its indicator-importance findings provide a baseline against which FinSight's SHAP results can be compared.

3.3 Literature Review Summary Table

Table 3.1 contains a compact summary of the literature reviews we carried out.

Table 3.1: Summary of the reviewed research studies.

Author	Method	Results	Limitations
Jiang and Zeng [1]	FinBERT sentiment + LSTM; compared with BERT, LSTM, ARIMA	Sentiment improves anticipation of market fluctuations	English news only; no regime; no explainability

Author	**Method**	**Results**	**Limitations**
Kim et al. [2]	FinBERT on summarised NYT news + LSTM, S&P 500	Summarised text is effective for sentiment-based forecasting	Single US index and source; no regime or XAI
Srijiranon et al. [3]	PCA-EMD-LSTM with FinBERT sentiment, Thai market, 2018–2022	Best performance versus baselines; sentiment improves LSTM	No regime handling or interpretability
Kasture and Shirsath [4]	Hybrid RNN-LSTM with sentiment scores, BSE Sensex; compared with RFR, SVR, LSTM	Sentiment enriches predictive capability	Single Indian index; no regime or XAI
Dhokane and Agarwal [5]	LSTM with EMA and RSI plus OHLCV, NSE India data	Improved RMSE, MAPE and R^2 with indicators	Price-only; no text, regime or XAI
Dogra et al. [6]	DistilBERT vs TF-IDF for 3-class sentiment of Indian banking and global news	Evaluates contextual transformer sentiment classification	Banking domain; not linked to forecasting
Chari et al. [7]	GARCH, regression, dictionary-based aggregate news sentiment, India	Short-lived sentiment effect; business and politics categories matter	Dictionary method; only days of large movement
Muhammad et al. [8]	Deep model for 5 trend classes; SHAP and LIME; 4 datasets	94.9% average accuracy vs 85.7% RF, 60.07% SVM, 52.45% LR	Trend classification; global explanations only
Wallapure Manikrao et al. [9]	Deep neural networks (incl. LSTM) with SHAP and LIME	Interpretable stock prediction without losing predictive power	Global, not regime-specific, explanations
Ruan and Jiang [10]	FinBERT headline sentiment + indicators + XGBoost + SHAP, S&P 500 stocks, 2018–2023	Beats technical-only and lexicon baselines; FinBERT features rank highly in SHAP	Tree model; US only; global explanations

Author	Method	Results	Limitations
Pagliaro [11]	Regime-aware LightGBM; rolling HMM; walk-forward with purge; SHAP	54.2% mean accuracy over 51 stocks; regime-dependent logic; Sharpe not DSR-significant	US equities; no sentiment; not a sequence model
Tian et al. [12]	KF-MS-LSTM/GRU on CSI 300, S&P 500, Nikkei 225	Regime conditioning is the main source of improvement	No sentiment or XAI; no frontier market
Ampountolas [13]	K-Means, DBSCAN, GMM regime clustering, Jul 2022–Feb 2025	Distinct regimes identified around the 2024 US election	Descriptive; short window; US and crypto
Ghani and Ghani [14]	GARCH-MIDAS with EPU indices, combination forecasts, MCS tests	US EPU strongest predictor of Pakistan volatility; Pakistan EPU not significant	Volatility only; no deep learning or sentiment
Mehmood and Ali [15]	LSTM and BiLSTM with attention plus NLTK sentiment, two PSX banks	Sentiment lowers mean squared error	Lexicon sentiment; two stocks; no regime or XAI
Raza and Akhtar [16]	ANN, SVM, Random Forest with 27 technical indicators, PSX	Key indicators: %R, Momentum, Disparity 5	Technical data only; no sentiment, regime or XAI

3.4 Related Products and Applications

Besides academic work, investors on the PSX already use a number of commercial and public tools. This section reviews six of them, five built for the Pakistani market and one international charting platform that also covers the KSE-100. For each product we describe its main features and compare it with FinSight. The descriptions are based on the public websites and documentation of the vendors, as accessed in October 2026, and a feature that is not mentioned there is treated as not offered.

3.4.1 PSX Data Portal

The PSX Data Portal (`dps.psx.com.pk`) is the official market data website of the Pakistan Stock Exchange [17]. Its core data can be viewed for free and without registration [18]. It offers a market

watch with current prices, intraday and end-of-day historical data for each listed security, sector views, company announcements, downloadable daily files and a stock screener that filters listed companies by sector, price range and financial ratios.

Its strength is that it is the authoritative source for PSX data. It is, however, a data display tool: it provides no forecasts, no sentiment analysis and no market regime information. PSX also states that commercial use or redistribution of its data requires prior approval and a licence [19], which is a point FinSight must consider if it is deployed publicly. The prices used in this project come from an Investing.com export instead.

3.4.2 Sarmaaya.pk

Sarmaaya (`sarmaaya.pk`) is a Pakistani investment tracking platform [20, 21], run by Sarmaaya Financials, which PSX lists as an authorised redistributor of its market data [19]. It is available on the web and as Android and iOS applications. It brings several asset classes into one platform, namely stocks, mutual funds, commodities and forex, and lets investors follow their whole portfolio in real time. Its other features are a market overview, charting through a trading view page, analysis tools for each asset class including an economy section, subscription plans with longer holding-period analytics, and training and expert-written articles that promote financial literacy.

The platform is strong in portfolio tracking and education. Its public description does not mention news sentiment scoring, price forecasting or explanations of predictions, so it describes the market as it is rather than where it may go.

3.4.3 Capital Stake Analytics Platform

Capital Stake, a Lahore-based fintech company launched in 2014, describes its platform as an end-to-end financial analytics platform that covers all companies listed on the PSX. A 2017 joint venture with PSX was reported to embed its analytics in the trading terminals used by more than 150 brokers [22]. The analytics portal that brokers offer to their clients includes real-time and historical data, an indices dashboard, top gainers and losers, financial statements and their history, live company announcements, economic indicators (for example GDP, the policy rate and the dollar–rupee rate), a market heat map, advanced technical charting, sector trackers, a news feed, a payout calendar and foreign and local investor portfolio flows [23].

This is the most complete analytics tool of those reviewed, and it already combines price, fundamental and macroeconomic indicators. Its news is shown as a feed that the user reads, and no sentiment score, forecast or regime label is described.

3.4.4 Mettis Global

Mettis Global is a Karachi-based provider of financial data, real-time news and analytical tools for Pakistan's financial markets [24], and is also an authorised redistributor of PSX data [19]. Its products include MG Link (news), MG Trader, MG FX, MG Portfolio Management, MG Rate Sheet and MG APIs, and its solutions are organised by job role, such as buy-side and sell-side equity, corporate treasury, fixed income and bank risk management, with separate offerings for students and universities.

Mettis combines local news and data and offers programmatic access through APIs, which is similar in delivery to the REST API planned for FinSight. Its public description does not mention automated sentiment scoring of news, forecasting models or explainability.

3.4.5 UpInvest

UpInvest (`upinvest.pk`) is a portfolio tracking platform for PSX investors, offered as a web and mobile application [25]. It tracks daily profit and loss, total portfolio performance, dividends and payouts in real time, calculates profit and loss with the FIFO method, and compares portfolio performance with the KSE-100 index.

The platform is useful for monitoring past performance and dividends. It is backward-looking and gives no forecasts, sentiment or market state, so it addresses a different need than FinSight.

3.4.6 TradingView

TradingView is an international web charting platform that also carries the KSE-100 index [26]. Its KSE-100 page shows a technical analysis summary that combines oscillators and moving averages, such as the RSI, MACD and exponential and simple moving averages, into an overall signal from strong sell to strong buy. The platform also provides drawing tools, a large library of technical indicators, and a scripting language with which users and the community create custom indicators and strategies. Some community indicators show sentiment, but those we found are based on prices or retail positioning for assets such as US stocks, forex and cryptocurrencies, and not on Pakistani news.

TradingView is strong in charting and technical signals, which are the same indicators that FinSight uses as features (RSI, MACD and EMA). Its signals are rule-based, and it provides no multi-horizon forecasts with uncertainty, no sentiment from Pakistani news and no explanation of what drives a prediction in each market regime.

3.4.7 Comparison with FinSight

Table 3.2 summarises the features of the products and the gap that FinSight addresses.

Table 3.2: Comparison of related products and applications with FinSight.

Product	Main features	Gap addressed by FinSight
PSX Data Portal [17]	Official market watch, intraday and historical data, sectors, announcements, downloads, stock screener	No forecasts, sentiment or regimes
Sarmaaya.pk [20]	Multi-asset portfolio tracking, market overview, charts, analysis tools, education content, web and mobile apps	Descriptive analytics only; no forecasts or explanations
Capital Stake [23]	Fundamentals, technical charting, economic indicators, heat map, announcements, news feed, broker terminal integration	News is not scored; no forecasts or regimes
Mettis Global [24]	Pakistani news, data, FX, portfolio tools and APIs for institutional users	No sentiment scoring, forecasting or explainability described
UpInvest [25]	Portfolio profit and loss, dividends, FIFO calculations, comparison with KSE-100	Backward-looking; no forecasts or sentiment
TradingView [26]	Charting, technical indicator signals, custom scripts for KSE-100 and other markets	Rule-based signals; no Pakistani news sentiment, uncertainty or regime-wise explanations

The products reviewed are strong at presenting market data, tracking portfolios, charting and delivering news. None of them, according to their public descriptions, scores Pakistani financial news with a sentiment model, labels the market regime, produces multi-horizon forecasts with confidence intervals, or explains a forecast. FinSight does not replace these tools: it adds a forward-looking, explainable layer on top of the data and news that they already provide.

3.5 Conclusion

The reviewed literature shows that sentiment-augmented LSTM forecasting is well established in global and Indian markets [1, 2, 3, 4, 5], that lightweight transformers and aggregate news sentiment are effective for South Asian financial text [6, 7], and that SHAP and LIME can make stock predictors interpretable [8, 9, 10]. Regime-aware studies indicate that conditioning on market state can improve forecasting [12] and change which features matter [11], and clustering offers a practical way to identify regimes [13]. For Pakistan, the studies we reviewed use uncertainty indices for volatility [14], lexicon-based sentiment on individual bank stocks [15], or technical indicators alone [16]. The existing products for Pakistani investors, such as the PSX Data Portal, Sarmaaya, Capital Stake, Mettis Global and UpInvest [17, 20, 23, 24, 25], and the international TradingView platform [26], provide data, charts, news and portfolio tracking but, as far as their public descriptions show, no sentiment-driven or regime-aware forecasts and no explanations. Within the literature and products reviewed, we did not identify a KSE-100 forecasting study or tool that combines Pakistan-specific news sentiment, regime-conditioned forecasting and per-regime explainability, nor a labelled Pakistani financial news sentiment dataset. FinSight is designed to address these gaps in a single pipeline and, in the next phase, to deliver the results through a web application.

Chapter 4 Software Requirement Specifications

This chapter presents the software requirements of FinSight. It includes the list of features, the functional and non-functional requirements, quality attributes, assumptions, use cases, hardware and software requirements, graphical user interface, database design and the risk analysis of the project. The requirements cover the complete system: the backend pipeline is implemented in the current phase, and the web application and REST API will be developed in the next phase.

4.1 List of Features

The major features of our system are as follows.

- A live KSE-100 dashboard with price data and technical indicators.
- Multi-horizon (3, 7 and 14 day) KSE-100 forecasts with confidence intervals.
- A daily sentiment feed built from Pakistani financial news headlines.
- Detection of market regimes (bull, bear and sideways) from KSE-100 price behaviour.
- An interactive explainability panel showing SHAP feature importance, computed separately for each market regime.
- A REST API that exposes the forecasts, sentiment scores and explanations to other applications.

4.2 Functional Requirements

This section describes the functional requirements of our system for its users.

4.2.1 Functional Requirements for Visitor

- The system should allow the visitor to view the live KSE-100 dashboard.
- The system should allow the visitor to view the technical indicators (EMA, RSI and MACD) of the KSE-100 index.
- The system should allow the visitor to view 3-day, 7-day and 14-day forecasts of the KSE-100 index.
- The system should display confidence intervals with each forecast.
- The system should allow the visitor to view the daily sentiment feed.
- The system should display the sentiment label and confidence of each news headline in the feed.

- The system should allow the visitor to view the current market regime.
- The system should allow the visitor to view the SHAP explanation of a forecast.
- The system should allow the visitor to view the SHAP feature importance separately for each market regime.

4.2.2 Functional Requirements for System

- The system should collect KSE-100 price data, Pakistani financial news headlines and the Pakistan EPU index.
- The system should compute the technical indicators (EMA-12, EMA-26, RSI-14 and MACD) from the KSE-100 price data.
- The system should classify the sentiment of each headline using the fine-tuned DistilBERT model.
- The system should aggregate the headline sentiments into daily sentiment scores.
- The system should detect the market regime (bull, bear or sideways) from KSE-100 price behaviour using clustering.
- The system should generate forecasts for 3, 7 and 14 days using the hybrid LSTM model.
- The system should generate SHAP explanations of the forecasts for each market regime.
- The system should expose the forecasts, sentiment scores and explanations through a REST API.

4.3 Quality Attributes

Quality attributes help in assessing the quality of the software. The quality attributes of our system are as follows.

- The system shall be easy and intuitive to use.
- The system should perform its functionalities with minimal errors and high accuracy.
- The system shall be easy to modify and upgrade through modules that are easy to maintain.
- The system shall make its predictions explainable.
- The system shall be reliable in collecting, storing and displaying data.

4.4 Non-Functional Requirements

This section describes the non-functional requirements of our system, focusing on factors such as performance, usability and extensibility.

- The system shall load data and display forecasts and explanations within an acceptable time.
- The system shall be responsive and usable on different screen sizes.
- The system shall present forecasts and explanations in a form that is easy for non-technical users to understand.
- The system shall support multiple visitors using the web application at the same time.
- The system shall allow new data sources or models to be added with minimal changes to the rest of the system.
- The system shall not execute trades, manage portfolios or provide personalized investment advice.

4.5 Assumptions

The following assumptions will be made when developing our system.

- Headlines from Dawn Business can be scraped throughout the project.
- KSE-100 daily OHLCV data from Investing.com (2015 to present) is available.
- The Pakistan EPU index is available to be used as the macroeconomic feature.
- The collected headlines can be manually annotated to create the labeled sentiment dataset.
- The system forecasts the KSE-100 index as a whole and not individual stocks.
- Sentiment data is limited to Dawn Business headlines; social media and forum sentiment is out of scope.

4.6 Use Cases

This section presents the use cases of the main functionalities offered by our system in Tables 4.1 to 4.4.

Table 4.1: Use case: View KSE-100 Dashboard.

View KSE-100 Dashboard
Actor: Visitor
The visitor opens the web application and views the live KSE-100 dashboard with the price data, technical indicators and current market regime.
The system must have KSE-100 data available.The dashboard is displayed to the visitor.None

Basic Flow			
Actor Action		**System Response**	
1	The visitor opens the dashboard page.	2	The system displays the live KSE-100 dashboard.
Alternative Flow			
1	The visitor opens the dashboard page when no data is available.	2-A	The system displays a message that data is not available.

Table 4.2: Use case: View Multi-Horizon Forecast.

View Multi-Horizon Forecast
Actor: Visitor
The visitor views the 3, 7 or 14 day forecast of the KSE-100 index with confidence intervals.
Forecasts must have been generated by the system.The selected forecast is displayed to the visitor.None

Basic Flow			
Actor Action		**System Response**	
1	The visitor opens the forecast page.	2	The system displays the forecast with its confidence interval.
3	The visitor selects a forecast horizon (3, 7 or 14 days).	4	The system displays the forecast and confidence interval for the selected horizon.
Alternative Flow			
1	The visitor opens the forecast page when no forecast is available.	2-A	The system displays a message that the forecast is not available.

Table 4.3: Use case: View Daily Sentiment Feed.

View Daily Sentiment Feed
Actor: Visitor
The visitor views the daily feed of Pakistani financial news headlines with their sentiment.
Headlines must have been collected and scored by the system.The sentiment feed is displayed to the visitor.None

Basic Flow			
Actor Action		**System Response**	
1	The visitor opens the sentiment feed page.	2	The system displays the headlines with their sentiment labels and the daily aggregated sentiment.
Alternative Flow			
1	The visitor opens the sentiment feed page when no headlines are available.	2-A	The system displays a message that no headlines are available.

Table 4.4: Use case: View Explanation by Market Regime.

View Explanation by Market Regime
Actor: Visitor
The visitor views the SHAP explanation showing which features drove a forecast, separately for each market regime. SHAP explanations must have been generated by the system.The selected explanation is displayed to the visitor.None

Basic Flow			
Actor Action		**System Response**	
1	The visitor opens the explainability panel.	2	The system displays the SHAP feature importance of the forecast.
3	The visitor selects a market regime (bull, bear or sideways).	4	The system displays the SHAP feature importance for the selected regime.
Alternative Flow			
1	The visitor opens the explainability panel when no explanation is available.	2-A	The system displays a message that the explanation is not available.

4.7 Hardware and Software Requirements

All the hardware and software requirements needed to develop and deploy the project are mentioned in this section.

4.7.1 Hardware Requirements

The hardware needed for developing and running the system is as follows.

- A computer with a GPU to aid faster fine-tuning of DistilBERT and training of the LSTM model.
- A server to host the web application.
- Sufficient storage for price data, headlines and forecasts.

4.7.2 Software Requirements

The software needed for developing the system is as follows.

- Libraries for fine-tuning DistilBERT, building the LSTM model, clustering and SHAP.
- A tool for scraping Dawn Business headlines.
- Investing.com as the source of KSE-100 historical data.
- A database to store the data used by the web application.
- A web framework for the web application.
- A framework for building the REST API.

4.8 Graphical User Interface

This section displays the GUI designs of the planned web application. The description of the GUI screens and the navigation flows are explained in the figure descriptions.

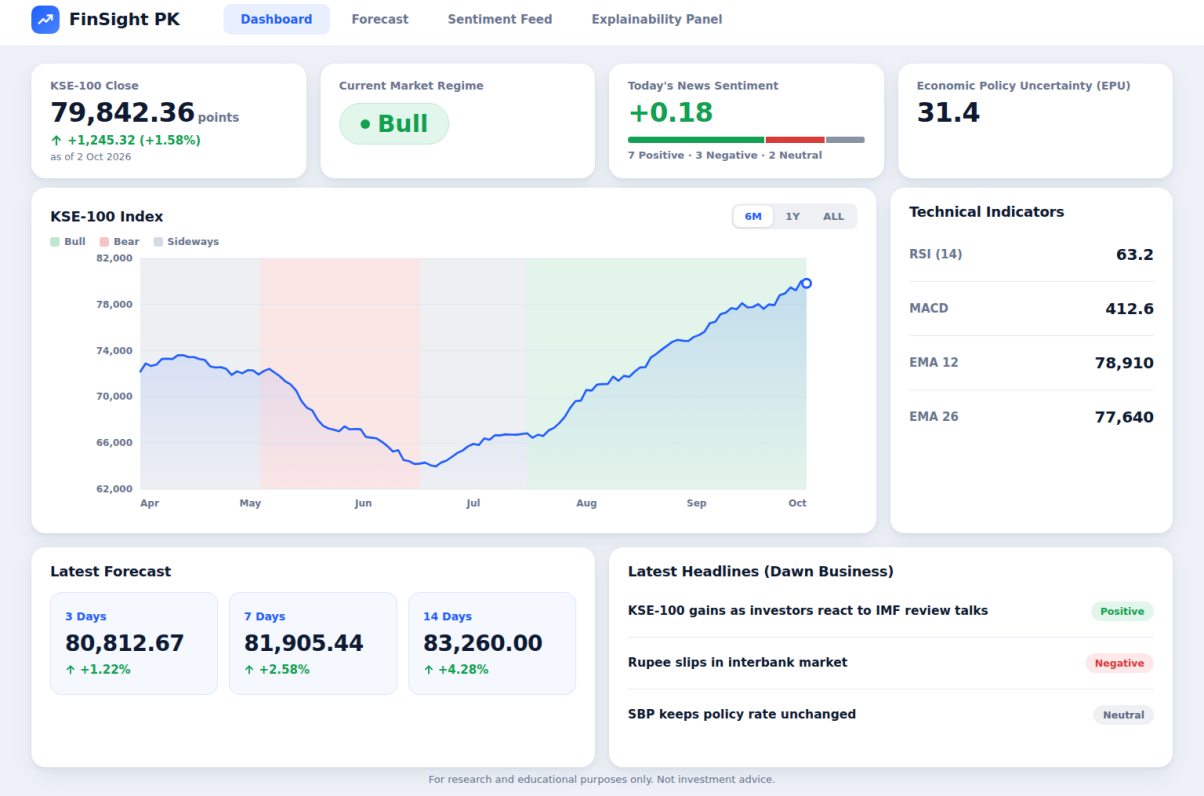


Figure 4.1: Dashboard screen showing live KSE-100 prices, indicators and market regime

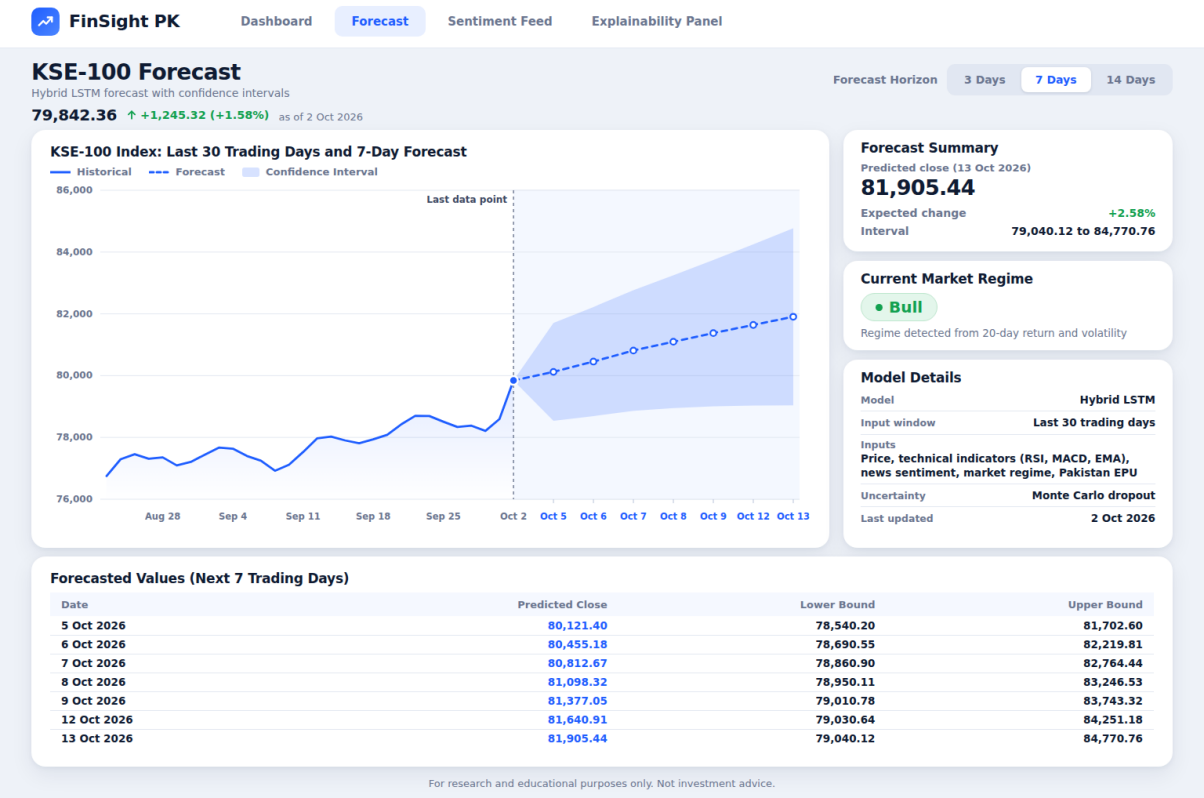


Figure 4.2: Forecast screen showing 3, 7 and 14 day forecasts with confidence intervals

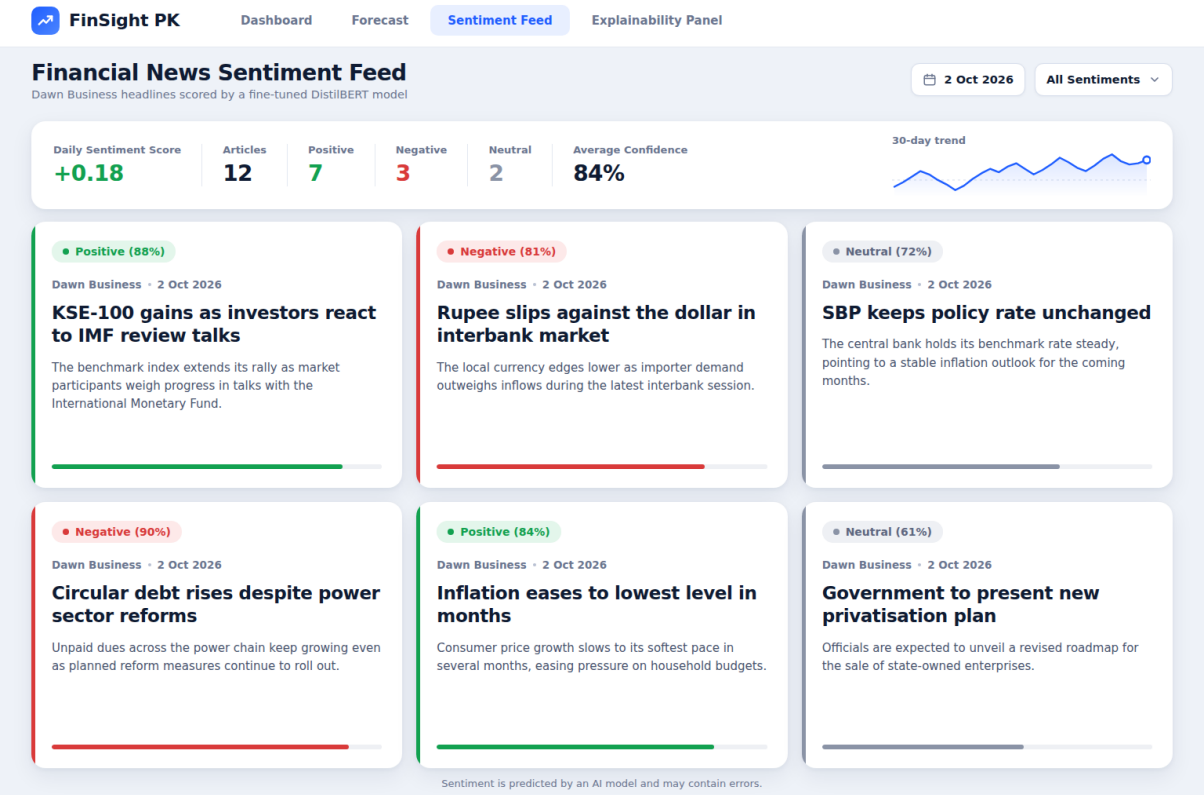


Figure 4.3: Sentiment feed screen showing headlines with sentiment labels

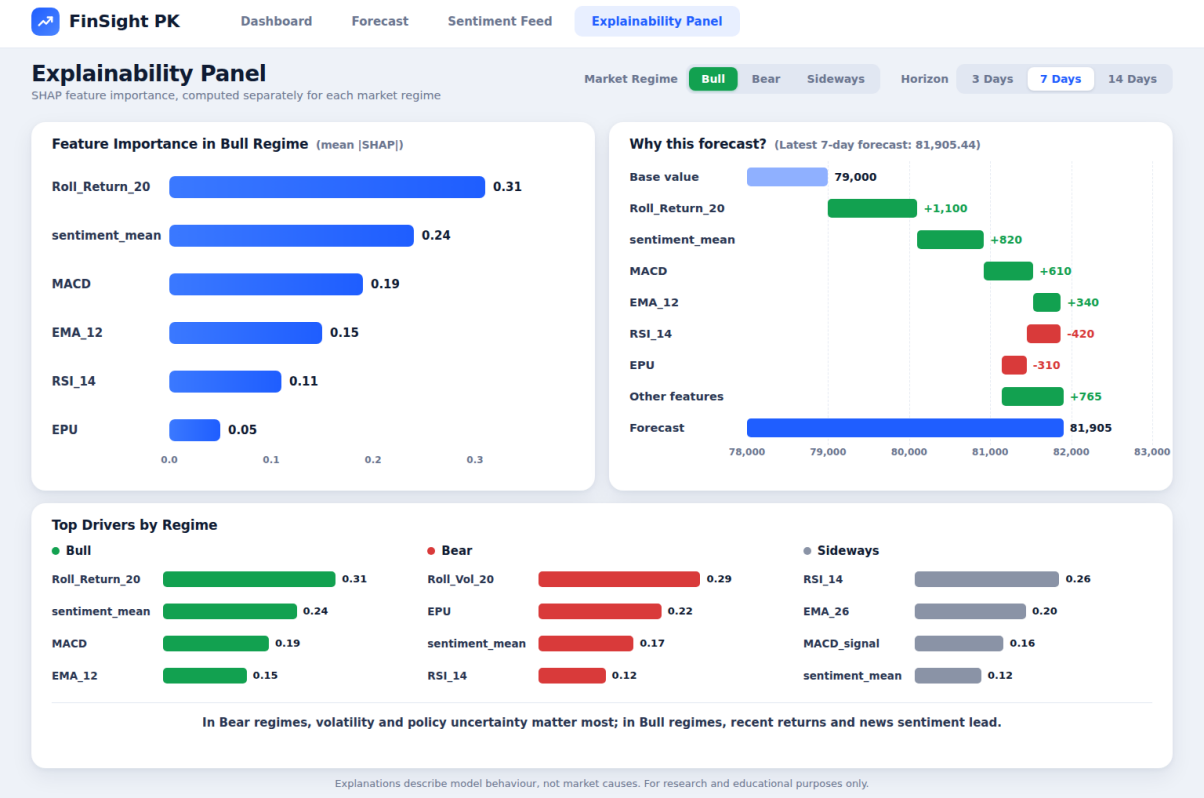


Figure 4.4: Explainability screen showing SHAP importance for each market regime

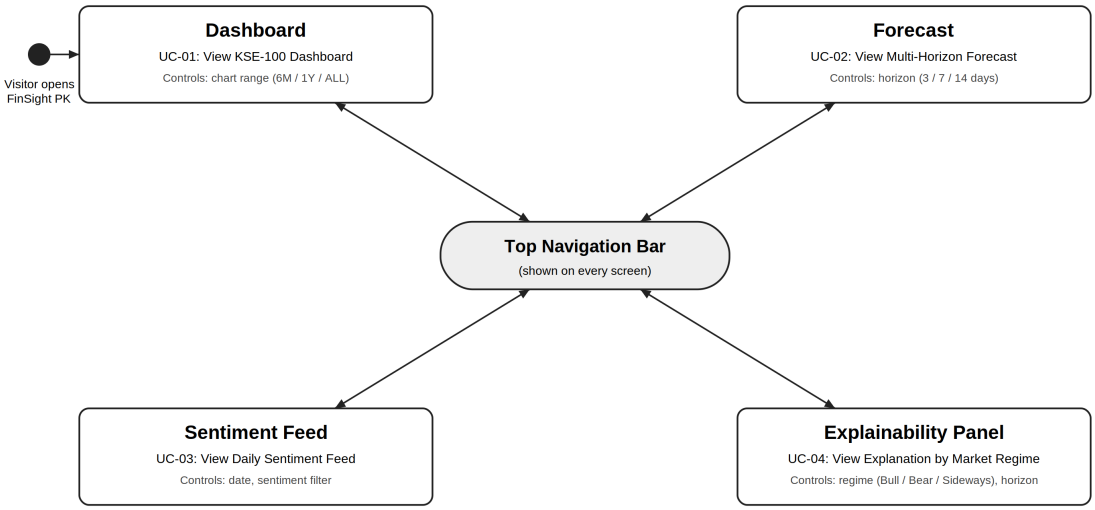


Figure 4.5: GUI navigation flow between the dashboard, forecast, sentiment feed and explainability screens

4.9 Database Design

This section presents the database design of our system through an ER diagram and a data dictionary describing the entities and their attributes.

4.9.1 ER Diagram

The Entity-Relationship (ER) diagram in Figure 4.6 represents the database design of our system. It includes the following entities.

- news_article: stores the scraped news articles (title, URL, publication date and the lead paragraph).
- article_sentiment: stores the predicted sentiment label, confidence and class probabilities of each article.
- daily_sentiment: stores the sentiment aggregated per day (sum, mean, counts and mean confidence).
- market_data: stores the daily KSE-100 OHLCV data and percentage change.
- technical_indicator: stores the EMA-12, EMA-26, RSI-14 and MACD values of each trading day.
- market_regime: stores the market regime (bull, bear or sideways) of each trading day along with the daily return, 20-day rolling volatility and 20-day rolling return.
- macro_epu: stores the Pakistan EPU index values.
- forecast: stores the 3, 7 and 14 day forecasts with confidence intervals.
- shap_value: stores the SHAP feature importance values for the forecasts and for each regime.

The relationships between the entities are as follows.

- Many news_article records are aggregated into one daily_sentiment record (N:1).
- Each news_article has one article_sentiment record (1:1).
- Each daily_sentiment record is aligned with one market_data record by date (1:1).
- Each macro_epu record is aligned with one market_data record by date (1:1).
- Each market_data record has one technical_indicator record (1:1).
- Each market_data record has one market_regime record (1:1).
- Many forecast records are based on one market_data record (N:1).
- Each forecast has many shap_value records (1:N).

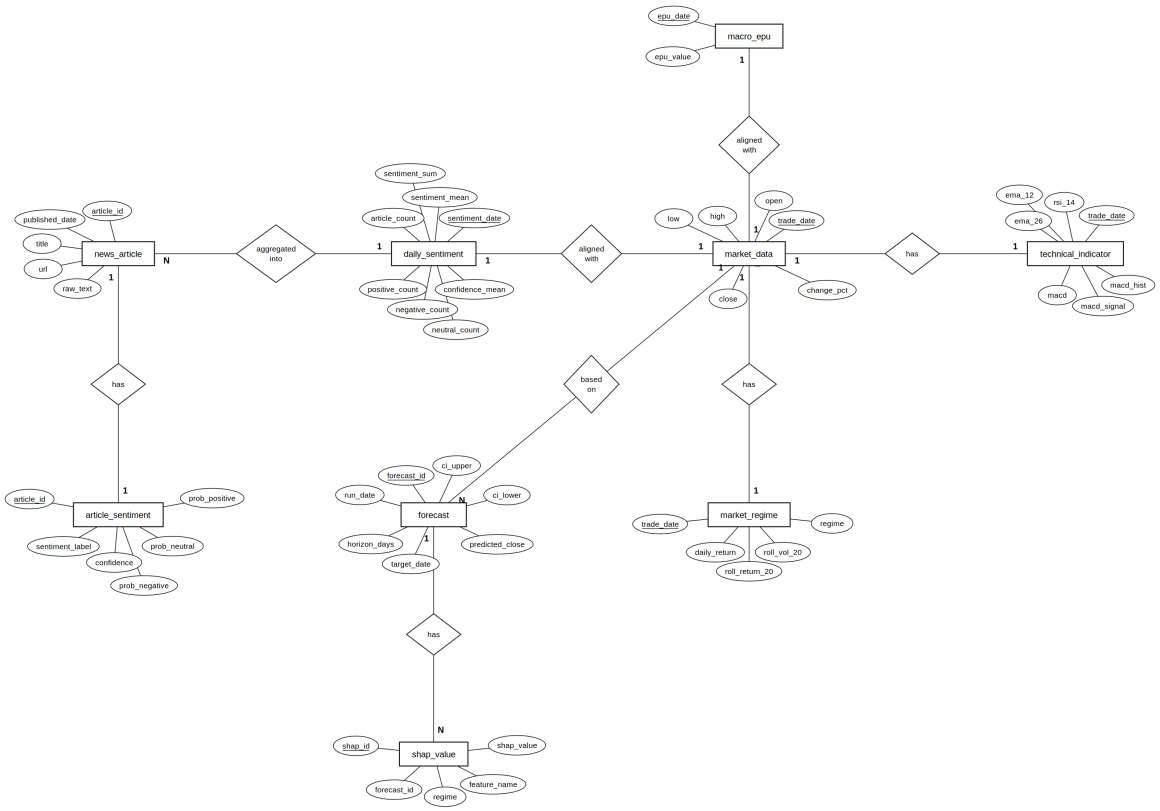


Figure 4.6: ER diagram. It describes the database design of our system.

4.9.2 Data Dictionary

The data dictionary in Tables 4.5 to 4.13 defines all the entities, their attributes and data types.

Table 4.5: Data dictionary: news_article.

Attribute	Type	Description
article_id (PK)	INTEGER	Unique identifier of the article.
published_date	DATE	Publication date.
title	TEXT	Headline of the article.
url	TEXT	Source URL of the article.
raw_text	TEXT	Lead paragraph of the article (the headline if the paragraph cannot be retrieved).

Table 4.6: Data dictionary: article_sentiment.

Attribute	Type	Description
article_id (PK, FK)	INTEGER	Article being scored.
sentiment_label	VARCHAR	Positive, negative or neutral.
confidence	DECIMAL	Confidence of the predicted label.
prob_positive	DECIMAL	Predicted probability of the positive class.
prob_neutral	DECIMAL	Predicted probability of the neutral class.
prob_negative	DECIMAL	Predicted probability of the negative class.

Table 4.7: Data dictionary: daily_sentiment.

Attribute	Type	Description
sentiment_date (PK)	DATE	Trading date of the aggregated score.
sentiment_sum	DECIMAL	Sum of the article sentiment scores of the day.
sentiment_mean	DECIMAL	Mean sentiment score of the day.
article_count	INTEGER	Number of articles aggregated.
positive_count	INTEGER	Number of positive articles.
negative_count	INTEGER	Number of negative articles.
neutral_count	INTEGER	Number of neutral articles.
confidence_mean	DECIMAL	Mean prediction confidence of the day.

Table 4.8: Data dictionary: market_data.

Attribute	Type	Description
trade_date (PK)	DATE	Trading date.
open, high, low, close	DECIMAL	Daily index values.
volume	BIGINT	Daily traded volume.
change_pct	DECIMAL	Percentage change of the index.

Table 4.9: Data dictionary: technical_indicator.

Attribute	Type	Description
trade_date (PK, FK)	DATE	Trading date.
ema_12	DECIMAL	12-day exponential moving average.
ema_26	DECIMAL	26-day exponential moving average.
rsi_14	DECIMAL	14-day relative strength index.
macd	DECIMAL	MACD line value.
macd_signal	DECIMAL	MACD signal line value.
macd_hist	DECIMAL	MACD histogram value.

Table 4.10: Data dictionary: market_regime.

Attribute	Type	Description
trade_date (PK, FK)	DATE	Date of the regime label.
regime	VARCHAR	Bull, bear or sideways.
daily_return	DECIMAL	Daily return of the index.
roll_vol_20	DECIMAL	20-day rolling volatility.
roll_return_20	DECIMAL	20-day rolling return (mean of the daily returns).

Table 4.11: Data dictionary: macro_epu.

Attribute	Type	Description
epu_date (PK)	DATE	Period of the value.
epu_value	DECIMAL	Pakistan EPU index value.

Table 4.12: Data dictionary: forecast.

Attribute	Type	Description
forecast_id (PK)	INTEGER	Unique identifier.
run_date	DATE	Date the forecast was generated.
horizon_days	INTEGER	3, 7 or 14.
target_date	DATE	Date being forecast.
predicted_close	DECIMAL	Forecast KSE-100 closing value.
ci_lower, ci_upper	DECIMAL	Limits of the confidence interval.

Table 4.13: Data dictionary: shap_value.

Attribute	Type	Description
shap_id (PK)	INTEGER	Unique identifier.
forecast_id (FK)	INTEGER	Forecast being explained.
regime	VARCHAR	Regime the importance applies to.
feature_name	VARCHAR	Input feature (for example sentiment mean or EPU).
shap_value	DECIMAL	SHAP importance value of the feature.

4.10 Risk Analysis

The following risks are associated with this project.

- Time constraint: the project involves many phases, namely dataset construction and manual annotation, model fine-tuning, regime detection, LSTM forecasting, SHAP explainability and web application development, so time needs to be managed carefully.
- Data availability: headlines are scraped from Dawn Business, so changes to the website or restrictions on scraping may affect data collection. KSE-100 prices are exported manually from Investing.com, so keeping the data up to date depends on a new export.
- Annotation effort: the sentiment dataset has to be annotated manually, which is time consuming.
- Performance: the regime-aware model may not improve over conventional non-regime-aware forecasting models, which is the question this project investigates.
- Macroeconomic feature: existing work on Pakistan reports that the Pakistan EPU index has no significant predictive power for stock market volatility [14], so this feature may add little.

4.11 Conclusion

To conclude, this chapter specified the software requirements of FinSight. It listed the major features of the system, namely the live KSE-100 dashboard, the 3, 7 and 14 day forecasts with confidence intervals, the daily sentiment feed, market regime detection and the regime-wise SHAP explanations. It then defined the functional requirements for the visitor and for the system, together with the quality attributes, non-functional requirements and assumptions that guide the development. The main functionalities were described through use cases, and the hardware and software needed to build and deploy the system were identified. The GUI screens and their navigation flow were presented, and the database was designed through an ER diagram and a data dictionary covering the market data, technical indicators, regimes, news articles and sentiment, EPU values, forecasts and SHAP values. Finally, the chapter analysed the main risks of the project, such as time constraints, data availability, annotation effort and the limited predictive power of the EPU index. These requirements form the basis of the following chapters.

Chapter 5 Proposed Approach and Methodology

This chapter describes the proposed approach of FinSight. The approach is a data-driven framework that combines market prices, news sentiment, macroeconomic uncertainty and market regimes to forecast the KSE-100 index over 3, 7 and 14 day horizons and to explain each forecast. The chapter covers the algorithmic and procedural details of the backend components that are not covered in the system design: data collection, sentiment modelling, feature engineering, regime detection, forecasting, uncertainty estimation, explainability and evaluation.

5.1 Overview of the Approach

The backend of FinSight is a sequential pipeline of seven stages. Each stage consumes the output of the previous stage and stores its results in the database described in Chapter 4.

- Stage 1 collects KSE-100 prices, Dawn Business news and the Pakistan EPU index.
- Stage 2 classifies the sentiment of every news headline using the fine-tuned DistilBERT model.
- Stage 3 aggregates headline sentiments into one daily score aligned to trading days.
- Stage 4 computes technical indicators and the market regime of each trading day.
- Stage 5 merges all signals into a single model-ready dataset.
- Stage 6 trains the hybrid LSTM and generates 3, 7 and 14 day forecasts with confidence intervals.
- Stage 7 computes SHAP explanations for each market regime.

Algorithm 1 summarises these seven stages in pseudocode, and Figure 5.1 shows the overall workflow from data collection to storage.

Data: KSE-100 OHLCV prices, Dawn Business headlines, Pakistan EPU index

Result: 3-, 7-, and 14-day forecasts with confidence intervals and SHAP explanations

Step 1: Data preparation;
collect KSE-100 prices, Dawn Business headlines, and Pakistan EPU data;
clean prices, drop volume, and compute daily returns;
carry the monthly EPU value to all trading days within each month;
Step 2: News sentiment processing;
manually review and label the headline dataset with FinBERT assistance;
fine-tune DistilBERT using the chronological 70:15:15 train/validation/test split with weighted cross-entropy;
foreach *headline* **do**
    predict p_{pos} and p_{neg} using the fine-tuned DistilBERT;
    compute sentiment score $s_i = p_{pos} - p_{neg}$;
end
aggregate headline sentiment scores by trading day using the next-trading-day rule;
Step 3: Market feature engineering;
compute technical indicators including EMA, RSI, and MACD;
compute 20-day return and volatility statistics;
assign each trading day to a bull, bear, or sideways regime using K-Means clustering;
Step 4: Dataset construction;
merge prices, technical indicators, daily sentiment, EPU, and regime on trading date;
one-hot encode the market regime;
perform a chronological train/validation/test split;
fit feature scaling using training data only;
Step 5: Forecasting;
construct 30-day sliding windows from the merged dataset;
train the hybrid LSTM forecasting model;
foreach *horizon* $h \in \{3, 7, 14\}$ **do**
    generate the h-day forecast;
    perform Monte Carlo dropout to estimate predictive uncertainty and confidence intervals;
end
Step 6: Evaluation and explanation;
evaluate forecasts against baseline models using RMSE, MAE, MAPE, and directional accuracy;
compute SHAP values for forecasts separately for each market regime;
Step 7: Storage;
store forecasts, confidence intervals, evaluation results, and SHAP explanations in PostgreSQL;

Algorithm 1: End-to-end backend pipeline of FinSight

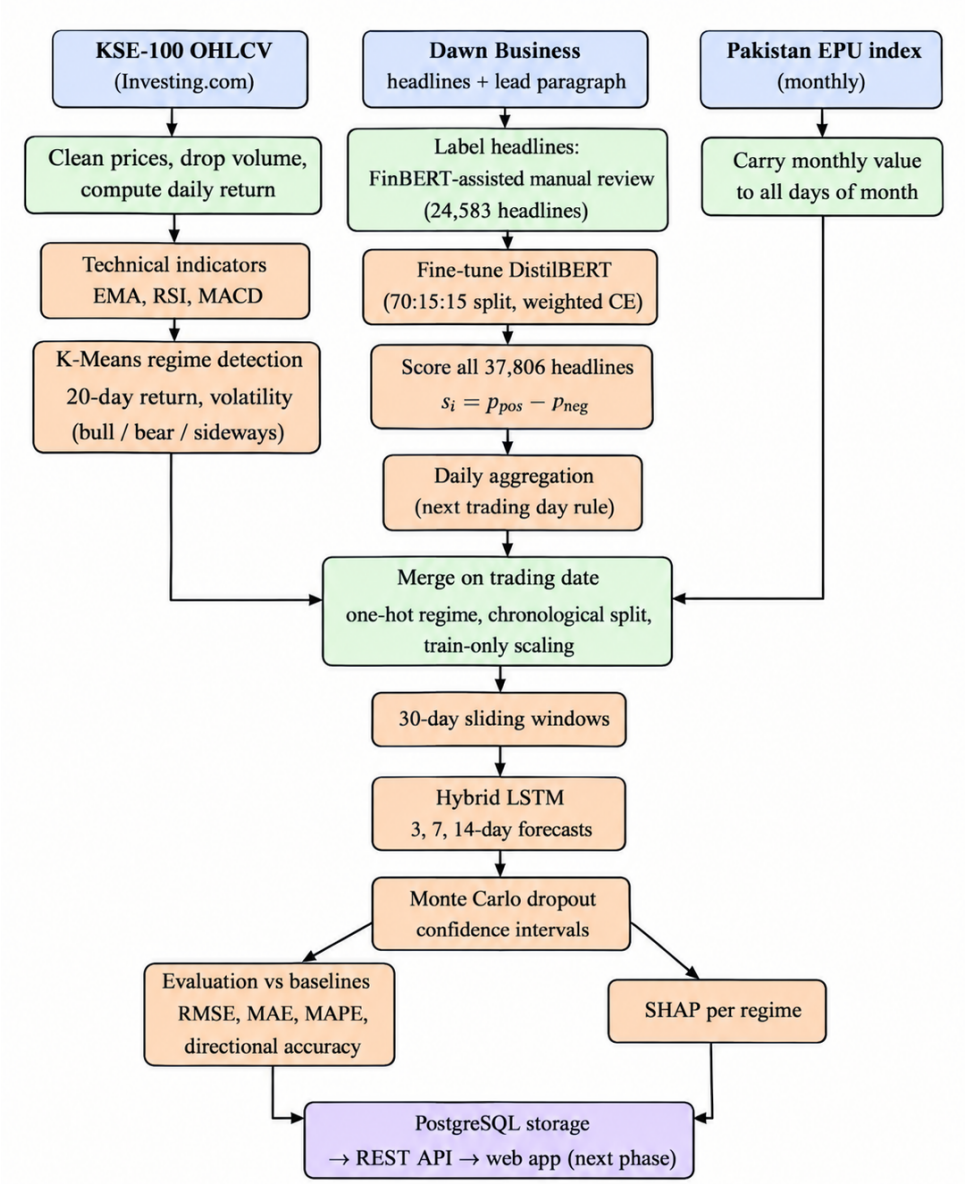


Figure 5.1: Proposed workflow of FinSight, from data collection to forecasts, explanations, and storage.

5.2 Data Collection and Preprocessing

5.2.1 KSE-100 Price Data

Daily OHLCV values and the daily percentage change of the KSE-100 index are taken from the historical data export of Investing.com. The export is restricted to a fixed window that starts in September 2014 and ends at the current date, so that every other dataset shares the same window and the first trading days of 2015 already have complete indicator values. Dates are parsed and sorted chronologically, price columns are converted to numbers, and volume values written with suffixes (K, M, B) are converted to plain numbers. Rows without a closing price are removed. The volume attribute is not used as a model feature because about 40% of its values are missing and the gaps are concentrated between 2019 and

2024 rather than being random, so imputing them would introduce bias.

5.2.2 News Headlines

Headlines are collected from the daily archive pages of the Dawn Business section, one page per calendar day from 2015 onwards. Only links to Dawn news articles whose publication date matches the archive date are kept. For every headline the scraper opens the article page and extracts the lead paragraph, which is the first paragraph longer than 60 characters after metadata lines such as the publication notice are skipped. If the paragraph cannot be retrieved after three attempts, the headline text itself is used instead. Requests are spaced by random delays and retried on failure to respect the source website. The scraper processes the archive month by month, appends the results of each month to a CSV file and skips dates that are already stored, so that an interrupted run resumes without repeating completed work. Duplicates are removed by URL. For each headline the title, URL, publication date and lead paragraph are stored.

5.2.3 Macroeconomic Data

The Pakistan EPU index is a monthly series, and no daily series exists for Pakistan. The EPU-4 value is used where available and the EPU-2 value is used as fallback for months in which EPU-4 is missing. The monthly value is carried over to every calendar day of its month, which aligns it with the daily KSE-100 data. Trading days before the start of the EPU series are dropped when the dataset is assembled.

5.3 Sentiment Analysis

5.3.1 Labelled Dataset Construction

A domain-specific labelled dataset is required because general financial sentiment models misread the direction of macroeconomic metrics such as deficits, inflation and circular debt. The lead paragraphs of the collected headlines are first scored automatically with FinBERT, and the predicted labels are then reviewed and corrected manually. The review sample is stratified over categories of known model failure, such as headlines containing bias-trigger terms, low-confidence predictions and metrics that are improving or worsening, so that every failure type is represented. The resulting labelled dataset contains 24,583 headlines, which are split into 17,208 training, 3,688 test and 3,687 validation headlines (70:15:15). Headlines whose sentiment cannot be resolved are removed.

5.3.2 Fine-Tuning DistilBERT

DistilBERT is fine-tuned for three-class classification (positive, neutral and negative) on the labelled dataset. The model input is the headline followed by its lead paragraph, truncated to 256 tokens. The loss is cross-entropy weighted by the inverse frequency of each class to compensate for class imbalance:

$$\mathscr{L} = -\sum_{i=1}^{N} w_{y_i} \log p_{i,y_i} \tag{5.1}$$

where p_{i,y_i} is the probability that the model assigns to the true class y_i of headline i and w_{y_i} is the weight of that class. The checkpoint with the best validation macro-F1 is retained and evaluated once on the held-out test set, where it reaches an accuracy of 82.7%. The retained model is then run over the whole collected corpus of 37,806 headlines. For each headline it outputs a label, a confidence value (the highest class probability) and the three class probabilities, which are stored in the `article_sentiment` entity.

5.3.3 Daily Sentiment Aggregation

The sentiment score of headline i is defined from its class probabilities as

$$s_i = p_{i,\text{pos}} - p_{i,\text{neg}} \tag{5.2}$$

so that $s_i \in [-1, 1]$. Using the probabilities instead of the label alone prevents low-confidence headlines from moving the daily average as much as confident ones. For a calendar day c with the set of headlines A_c, the mean score is

$$S_c = \frac{1}{|A_c|} \sum_{i \in A_c} s_i \tag{5.3}$$

News is published every day of the week while the market trades only on some days, so each calendar day is assigned to the next trading day on or after it. A weekend or holiday headline therefore informs the next session, which is the first one in which the market can react to it. The daily sentiment of a trading day d is the mean of S_c over all calendar days c assigned to it, and the sum of the scores, the number of headlines, the number of positive, negative and neutral headlines and the mean confidence are aggregated in the same way. A trading day with no headline receives zero for all sentiment features, because the absence of news is treated as neutral information. The sentiment of a trading day is combined with the closing data of the same day, so a forecast is issued after the market close.

5.4 Technical Indicators

The following indicators are computed directly from the closing price P_t using their standard definitions. The first rows of the series have undefined values while the indicators warm up, and they are dropped when the dataset is assembled. The exponential moving average with span n is

$$\mathrm{EMA}_t^{(n)} = \alpha P_t + (1-\alpha)\,\mathrm{EMA}_{t-1}^{(n)}, \qquad \alpha = \frac{2}{n+1} \tag{5.4}$$

The MACD line is the difference between the 12-day and the 26-day EMA, its signal line is the 9-day EMA of the MACD line, and the histogram is their difference:

$$\mathrm{MACD}_t = \mathrm{EMA}_t^{(12)} - \mathrm{EMA}_t^{(26)}, \qquad \mathrm{Hist}_t = \mathrm{MACD}_t - \mathrm{Signal}_t \tag{5.5}$$

The 14-day relative strength index uses Wilder's smoothing, an exponential average with $\alpha = 1/14$, for the gains and losses:

$$\mathrm{RSI}_t = 100 - \frac{100}{1+RS_t}, \qquad RS_t = \frac{\text{smoothed average gain}}{\text{smoothed average loss}} \tag{5.6}$$

5.5 Market Regime Detection

Market regimes are identified without labels, using clustering on two descriptors of recent market behaviour: the 20-day rolling return, defined as the mean of the daily returns of the last 20 days, and the 20-day rolling volatility, defined as their standard deviation. Both are computed from the daily return r_t:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \quad \sigma_t^{(20)} = \mathrm{std}\left(r_{t-19}, \ldots, r_t\right), \quad R_t^{(20)} = \frac{1}{20}\sum_{i=0}^{19} r_{t-i} \tag{5.7}$$

The two descriptors are standardised and grouped by K-Means with three clusters, with a fixed random seed and several random initialisations. The quality of the clustering is checked with the silhouette score. K-Means minimises the within-cluster sum of squares

$$J = \sum_{k=1}^{3} \sum_{\mathbf{x}_t \in C_k} \|\mathbf{x}_t - \mu_k\|^2 \tag{5.8}$$

where $\mathbf{x}_t$ is the standardised descriptor vector of day t and μ_k is the centroid of cluster C_k. Clusters are then named from their centroids: the cluster with the highest mean rolling return is labelled bull, the one with the lowest is labelled bear, and the remaining cluster is labelled sideways. The descriptors of a day use only past prices. The clustering, however, is fitted once on the whole history, so the centroids are

influenced by later data. The regime label is therefore a descriptive input of the model, and refitting the clusters in a walk-forward manner is a refinement that avoids this look-ahead.

5.6 Dataset Assembly

All signals are joined on the trading date into one dataset. Each row contains the OHLC prices, the technical indicators, the daily sentiment features, the EPU value and the regime label of one trading day. The regime is one-hot encoded before it is given to the model. Rows before the start of the EPU series and warm-up rows, in which the rolling indicators and the regime are not yet defined, are dropped. Features will be scaled using statistics computed on the training portion only, to avoid information leakage.

The dataset will be split chronologically into training, validation and test sets. Random shuffling is not used, because it would let the model see the future during training. The model input at day t is a sliding window of the previous 30 trading days of features.

5.7 Hybrid LSTM Forecasting Model

5.7.1 Model Structure

The forecaster is a hybrid model in the sense that it fuses heterogeneous inputs: price and indicator sequences, sentiment, macroeconomic uncertainty and the regime. The input window passes through LSTM layers, whose cell state is updated at each step by

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \tag{5.9}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \tag{5.10}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \tag{5.11}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \qquad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{5.12}$$

where $\mathbf{f}_t$, $\mathbf{i}_t$ and $\mathbf{o}_t$ are the forget, input and output gates. The final hidden state is passed to dense layers that output the forecast values of the three horizons.

5.7.2 Multi-Horizon Forecasting

The model produces the closing value of the index at 3, 7 and 14 trading days ahead from the same input window. Training minimises the mean squared error summed over the horizons:

$$\mathscr{L}_{\text{forecast}} = \frac{1}{N}\sum_{n=1}^{N}\sum_{h\in\{3,7,14\}}\left(\hat{y}_n^{(h)} - y_n^{(h)}\right)^2 \tag{5.13}$$

Training will use early stopping on the validation loss to limit overfitting.

5.7.3 Confidence Intervals

The confidence interval of each forecast is obtained with Monte Carlo dropout. Dropout layers are kept active at prediction time, and the same input window is passed through the network M times, each time with a different random dropout mask. This gives M stochastic forecasts $\hat{y}_1^{(h)},\ldots,\hat{y}_M^{(h)}$ for each horizon h. The point forecast is their mean, and the confidence interval is formed from their spread:

$$\bar{y}^{(h)} = \frac{1}{M}\sum_{m=1}^{M}\hat{y}_m^{(h)}, \qquad \left[\bar{y}^{(h)} - z\,\hat{\sigma}^{(h)},\ \bar{y}^{(h)} + z\,\hat{\sigma}^{(h)}\right] \tag{5.14}$$

where $\hat{\sigma}^{(h)}$ is the standard deviation of the M forecasts and z is the critical value of the chosen confidence level. Uncertainty typically grows with the horizon, so the interval widens from 3 to 14 days, which the dashboard displays to the visitor.

5.8 Regime-Wise Explainability

SHAP is used to attribute each forecast to its input features. The contribution of feature j is its Shapley value

$$\phi_j = \sum_{S\subseteq F\setminus\{j\}} \frac{|S|!\,(|F|-|S|-1)!}{|F|!}\left[f(S\cup\{j\}) - f(S)\right] \tag{5.15}$$

where F is the set of features and $f(S)$ is the model output when only the features in S are known. To expose how the drivers of the market change between regimes, the SHAP values are averaged in absolute terms separately over the days belonging to each regime:

$$I_j^{(k)} = \frac{1}{|D_k|}\sum_{t\in D_k}\left|\phi_j(t)\right| \tag{5.16}$$

where D_k is the set of days of regime k. The values $I_j^{(k)}$ are stored in the `shap_value` entity and shown in the explainability panel.

5.9 Evaluation Methodology

The regime-aware model is compared against baselines that do not use regimes, and against baselines that do not use sentiment. This ablation measures whether regime awareness and sentiment improve the forecasts, which is the main question of the project. Forecast accuracy is measured on the chronologically held-out test set using the root mean squared error, the mean absolute error and the mean absolute percentage error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{n}(y_n - \hat{y}_n)^2}, \quad \mathrm{MAE} = \frac{1}{N}\sum_{n}|y_n - \hat{y}_n|, \quad \mathrm{MAPE} = \frac{100}{N}\sum_{n}\left|\frac{y_n - \hat{y}_n}{y_n}\right| \tag{5.17}$$

together with the directional accuracy, the share of forecasts that predict the correct direction of movement. The sentiment model is evaluated by accuracy and macro-F1 on its held-out test set, and the Monte Carlo dropout confidence intervals are evaluated by their empirical coverage on the test set.

5.10 Conclusion

This chapter described the methodology of FinSight. News sentiment is obtained from a DistilBERT model fine-tuned on a manually corrected, domain-specific dataset and is aggregated into a daily score. Technical indicators and clustering-based market regimes are computed from prices, and all signals are merged into one dataset that is split chronologically. A hybrid LSTM with a 30-day input window forecasts the index at 3, 7 and 14 days with Monte Carlo dropout confidence intervals, and SHAP explains each forecast separately for every regime. The evaluation compares the regime-aware model with non-regime-aware baselines to answer the research question of the project.

Chapter 6 High-Level and Low-Level Design

This chapter presents the design of FinSight. It gives an overview of the system, the considerations that shaped the design, the layered system architecture with its subsystems, the architectural strategies and the domain model. It ends with the policies and tactics that apply to the details of the implementation.

6.1 System Overview

FinSight is a web-based forecasting and explanation system for the KSE-100 index. In the current phase the backend pipeline is implemented, while the REST API and the web frontend are designed in this chapter and will be developed in the next phase. A backend machine learning pipeline collects KSE-100 prices, Dawn Business news and the Pakistan EPU index, converts them into features (news sentiment, technical indicators and market regime) and trains a hybrid LSTM model that forecasts the index 3, 7 and 14 days ahead with confidence intervals. The forecasts, together with SHAP explanations computed separately for each market regime, are stored in a database and served to visitors through a REST API and a web frontend.

The design follows a layered organisation. External data sources feed a data collection layer, whose output is processed by the ML pipeline. The pipeline writes its results to the storage layer, and the application layer reads from storage and presents the results to the visitor through four screens: the dashboard, the forecast, the sentiment feed and the explainability panel. The visitor only reads information; the system has no user accounts, trading functions or personalized advice.

6.2 Design Considerations

6.2.1 Assumptions and Dependencies

The design depends on the following.

- Dawn Business headlines remain scrapable, and the page structure of the website is stable enough for the scraper to keep working.
- KSE-100 daily OHLCV data from Investing.com is available from 2015 to the present.
- The Pakistan EPU index is published monthly and its monthly value is carried over to every day of the month.
- The model is trained and the explanations are computed in Python on a machine with a GPU, while the web application is hosted on a server.

- Visitors use a standard web browser on a desktop or mobile device and need no technical background.
- The system forecasts the KSE-100 index as a whole, not individual stocks, and uses only Dawn Business news as sentiment source.
- New data sources or models may be added later, so each pipeline stage is kept independent of the others.

6.2.2 General Constraints

The following constraints affect the design.

- Data repository: all data used by the web application, namely articles, prices, indicators, regimes, sentiment, forecasts and SHAP values, must be stored in one relational database so that the application layer never depends on the ML pipeline at request time.
- Performance: forecast and explanation computation (including repeated Monte Carlo dropout passes and SHAP) is expensive, so it cannot run when a visitor opens a page. The results must be precomputed and only read at request time.
- Interface: the frontend communicates with the backend only through the REST API.
- Concurrency: multiple visitors must be served at the same time, which is possible because requests only read precomputed data.
- Data volume: the 30-day input window and the three horizons fix the shape of the model input and output, and the daily frequency of the data keeps storage requirements small.
- Scraping: data collection must respect the limits of the source website and must be resumable after an interruption.
- Scope: the system must not execute trades, manage portfolios or give personalized investment advice, so no user financial data is handled.

6.2.3 Goals and Guidelines

The design is guided by the following goals.

- Explainability: every forecast must be traceable to its input features, which is why a SHAP explainer is a first-class component and explanations are computed for each regime.
- Modularity: each component has one responsibility and communicates through well-defined inputs and outputs, so a component such as the sentiment model can be replaced without changing

the rest of the system.

- Separation of training and serving: heavy computation is done offline in the ML pipeline, while the application layer stays light and fast.
- Simplicity for the visitor: the screens present forecasts and explanations in a form that non-technical users can understand.
- Reliability of data: collected and computed data is persisted before it is shown, so a failure in one stage does not leave the web application without data.

6.2.4 Development Methods

The system was developed in an iterative and incremental manner. The data pipeline, the sentiment model, the regime detector, the forecasting model and the explainer were built and validated one after another, and each stage's output was checked before it became the input of the next stage. The web application was designed in parallel around the stored outputs. A strictly sequential (waterfall) approach was not used because the performance of the regime-aware model on real data was not known in advance, and the earlier stages, such as the sentiment model, had to be refined after analysing their errors.

6.3 System Architecture

The system is organised into five layers, as shown in Figure 6.1.

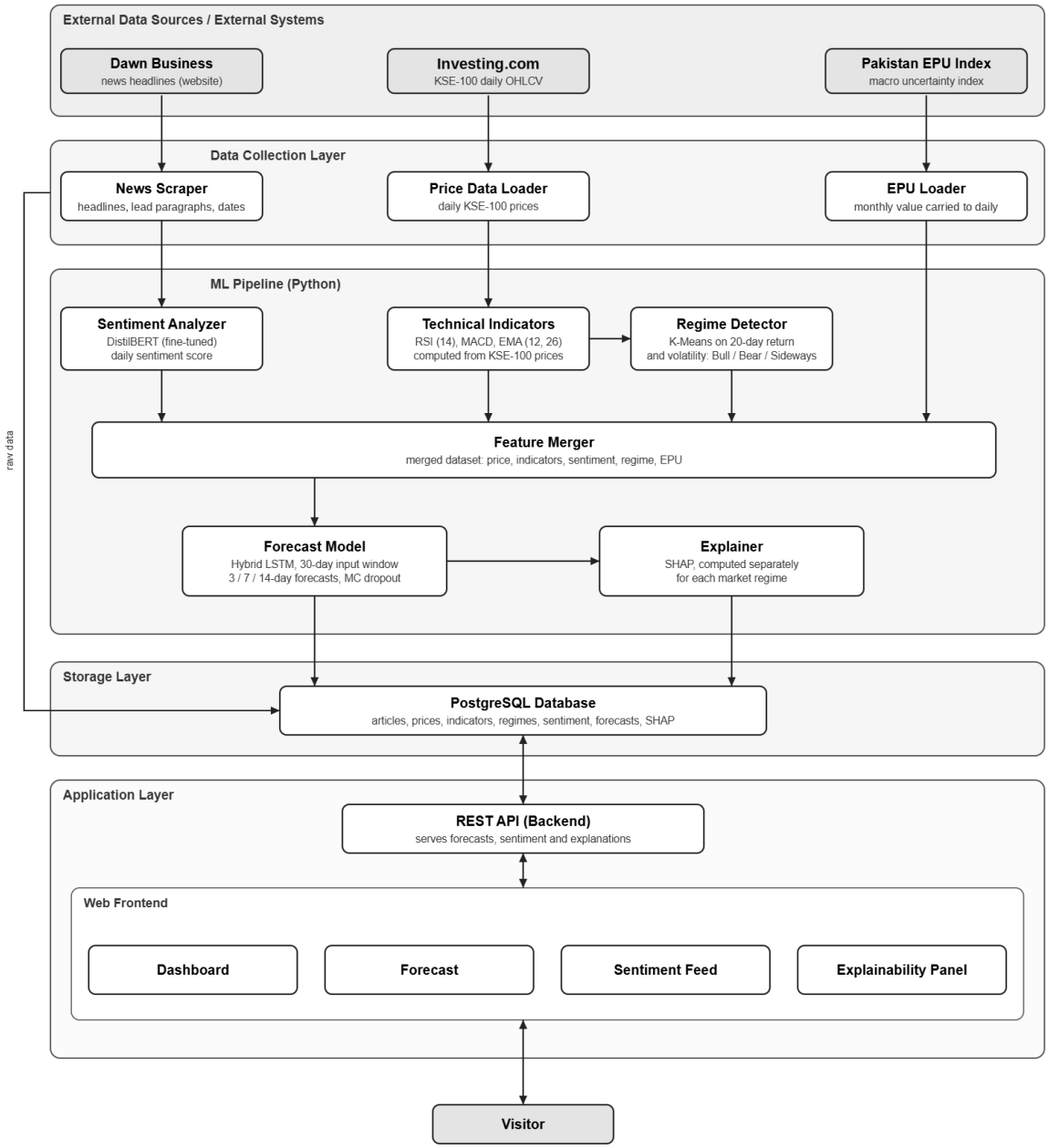


Figure 6.1: System architecture of FinSight showing the data, ML pipeline, storage and application layers

The responsibilities of each layer are as follows.

- External data sources: Dawn Business (news headlines), Investing.com (KSE-100 daily OHLCV) and the Pakistan EPU index (macroeconomic uncertainty). These are outside the system and are only read.
- Data collection layer: the news scraper collects headlines, lead paragraphs and dates, the price data loader reads the daily KSE-100 prices, and the EPU loader carries the monthly index over to daily frequency. The raw news data is also written directly to the database.

- ML pipeline: a Python pipeline that turns raw data into features, forecasts and explanations. Its components are described in the next subsection.
- Storage layer: a PostgreSQL database that holds articles, prices, indicators, regimes, sentiment, forecasts and SHAP values, following the schema of Chapter 4.
- Application layer: a REST API that serves forecasts, sentiment and explanations from the database, and a web frontend with the dashboard, forecast, sentiment feed and explainability panel screens that the visitor uses.

The layers collaborate in one direction for data and in two directions for requests. Data flows downwards from the sources through the pipeline into the database. At request time the visitor's browser calls the REST API, which reads the database and returns the precomputed results to the frontend.

This decomposition was chosen because it isolates the parts that change for different reasons: the scrapers change when a source changes, the ML pipeline changes when the models change, and the application layer changes when the user interface changes. An alternative design, in which the web application would run the models on demand, was rejected because of its latency and because it would couple the user interface to the model code.

6.3.1 Subsystem Architecture

The ML pipeline is the most complex subsystem and is divided into the following components.

- Sentiment analyzer: classifies each headline (with its lead paragraph) using the fine-tuned DistilBERT model and produces the daily sentiment score.
- Technical indicators: computes RSI (14), MACD and EMA (12, 26) from the KSE-100 prices.
- Regime detector: applies K-Means to the 20-day return and volatility derived from the prices and labels each trading day as bull, bear or sideways.
- Feature merger: joins prices, indicators, daily sentiment, regime and EPU into one dataset aligned on the trading date.
- Forecast model: a hybrid LSTM that takes a 30-day input window and outputs the 3, 7 and 14 day forecasts, with confidence intervals obtained by Monte Carlo dropout.
- Explainer: computes SHAP values of the forecasts separately for each market regime.

The technical indicators feed the regime detector, because the regime is derived from price behaviour. The sentiment analyzer, the technical indicators, the regime detector and the EPU loader all feed the feature merger, and the merged dataset is the only input of the forecast model. The explainer takes the

trained forecast model as its input, so the explanations always correspond to the forecasts that are shown. The forecast model and the explainer write their outputs to the database.

The application layer has two subsystems. The REST API exposes read operations for forecasts, sentiment and explanations. The web frontend consists of four screens, each backed by one group of API calls: the dashboard shows prices and indicators, the forecast screen shows the three horizons with confidence intervals, the sentiment feed shows the scored headlines, and the explainability panel shows SHAP importance for each regime.

6.4 Architectural Strategies

6.4.1 Layered Architecture

The system is built as layers in which each layer only uses the layer directly below or above it. This keeps the data collection, the ML pipeline, storage and presentation independent. The alternative of a single monolithic application was rejected because a change in a scraper or model would affect the whole system.

6.4.2 Offline Pipeline and Precomputed Results

All expensive computation, namely sentiment classification, model inference with Monte Carlo dropout and SHAP, runs in the ML pipeline and its results are stored. The web application only reads stored results. This gives fast page loads, supports many simultaneous visitors and keeps the web server independent of the GPU.

6.4.3 Relational Database as the Single Data Store

PostgreSQL is used as the single store for raw and derived data. A relational schema fits the strongly structured, date-keyed data of the system and its one-to-one and one-to-many relationships, and it allows the REST API to query results with simple joins. Keeping raw articles in the database as well as derived results supports traceability from a displayed forecast back to its inputs.

6.4.4 REST API Between Backend and Frontend

The frontend communicates with the backend only through the REST API. This decouples the user interface from the database and the pipeline, so the frontend can change without touching them, and the API can be reused by other clients in the future.

6.4.5 Uncertainty Through Monte Carlo Dropout

Confidence intervals are obtained by repeating the prediction with dropout active. This reuses the trained model and adds uncertainty without a second model. The alternative of training separate quantile models for each horizon was rejected because it would multiply training cost.

6.4.6 Regime-Wise Explainability

SHAP values are aggregated separately for the bull, bear and sideways regimes. This lets the visitor see how the importance of features such as sentiment and EPU changes between market conditions, which is the main explainability goal of the project.

6.4.7 Error Detection and Recovery

The news scraper saves its results after each month and skips dates that are already stored, so that an interrupted run resumes where it stopped. When a data source is unavailable or a stage fails, the application keeps serving the last results stored in the database, and when no data exists the screens show a message that the data is not available.

6.4.8 Future Extension

Because each pipeline stage has a defined input and output, new data sources (for example additional news outlets) or new models can be added by replacing or inserting a component, with little change to the rest of the system.

6.5 Domain Model/Class Diagram

The class diagram in Figure 6.2 represents the structure of the system. It contains two kinds of classes: processing classes that implement the backend components of the ML pipeline, and data classes that hold the stored entities and correspond to the tables of the database in Chapter 4.

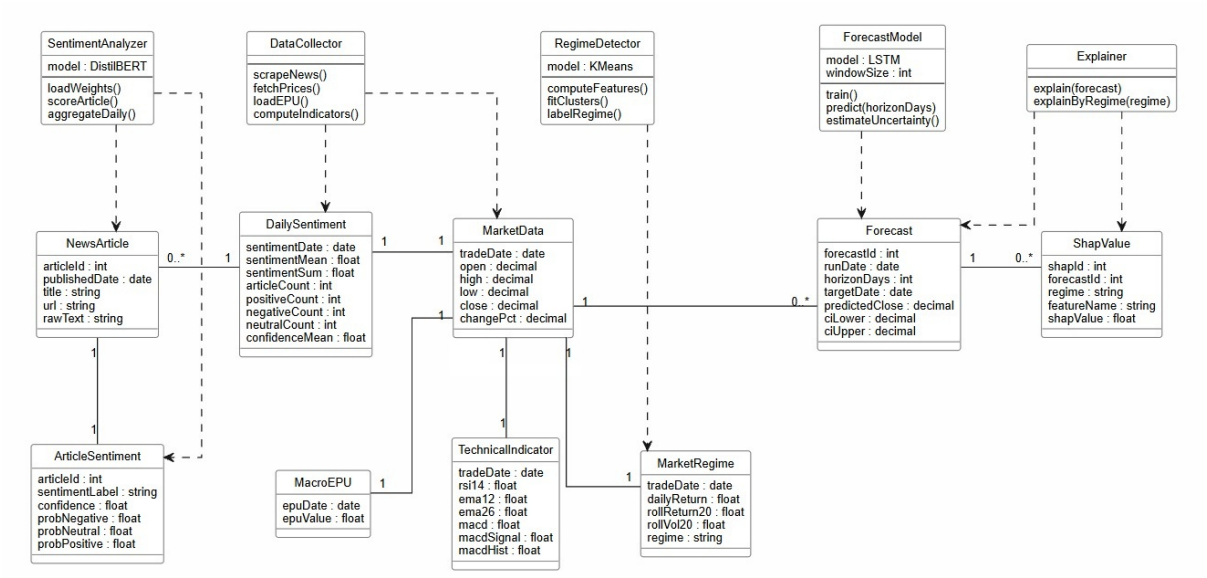


Figure 6.2: Class diagram of FinSight showing the backend processing classes and their data classes

6.5.1 Processing Classes

The processing classes correspond to the components of the architecture in Figure 6.1.

- SentimentAnalyzer: holds the DistilBERT model. It loads the fine-tuned weights (loadWeights), scores each article (scoreArticle) and aggregates the article scores into the daily sentiment (aggregateDaily). It creates ArticleSentiment objects and DailySentiment objects.
- DataCollector: collects the raw data and prepares the market features. It scrapes the news (scrapeNews), fetches the KSE-100 prices (fetchPrices), loads the EPU index (loadEPU) and computes the technical indicators (computeIndicators). It populates NewsArticle, MarketData, MacroEPU and TechnicalIndicator. In the architecture diagram the indicator computation is shown as a separate component of the ML pipeline, while in the class diagram it is an operation of DataCollector.
- RegimeDetector: holds the K-Means model. It computes the clustering features (computeFeatures), fits the clusters (fitClusters) and labels each day as bull, bear or sideways (labelRegime). It produces MarketRegime objects.
- ForecastModel: holds the LSTM model and its window size. It trains the model (train), predicts the forecast for a given horizon in days (predict) and estimates the uncertainty of the prediction (estimateUncertainty). It produces Forecast objects.
- Explainer: explains a forecast (explain) and explains it separately for a given market regime (explainByRegime). It reads Forecast objects and produces ShapValue objects.

6.5.2 Data Classes and Relationships

The data classes and their relationships are as follows.

- NewsArticle stores the scraped article (identifier, publication date, title, URL and the lead paragraph as raw text). Each article has exactly one ArticleSentiment (1:1), which stores the predicted label, the confidence and the three class probabilities.
- Many NewsArticle objects (0..*) are aggregated into one DailySentiment (1), which stores the daily sum, mean, article counts, label counts and mean confidence.
- Each DailySentiment is aligned with one MarketData object (1:1) through the trading date. MarketData stores the OHLC prices and the percentage change.
- Each MarketData object is related to exactly one MacroEPU object (1:1), one TechnicalIndicator object (1:1) and one MarketRegime object (1:1), all through the trading date. TechnicalIndicator stores RSI, the two EMAs and the MACD values, and MarketRegime stores the daily return, the 20-day rolling return, the 20-day rolling volatility and the regime label.
- One MarketData object is related to many Forecast objects (0..*). A Forecast stores the run date, the horizon in days, the target date, the predicted close and the lower and upper limits of the confidence interval.
- One Forecast has many ShapValue objects (0..*). A ShapValue stores the regime, the feature name and the SHAP value.

The dashed arrows in the diagram are dependencies, which show which data classes each processing class creates or uses. The solid lines are associations with their multiplicities, and they match the relationships of the ER diagram in Chapter 4.

6.6 Policies and Tactics

6.6.1 Choice of Technology

Python is used for the whole data and ML pipeline because of its libraries for scraping, transformers, LSTM models, clustering and SHAP. PostgreSQL is used for storage, and a REST API with a web frontend is used for the application layer. DistilBERT was chosen over a larger model because it is lighter to fine-tune and run, and over FinBERT because fine-tuning FinBERT on labels that largely come from its own predictions would be circular.

6.6.2 Data Handling Policy

Features and the data split follow chronological order, and scaling statistics will be computed on the training portion only, so that no future information reaches the model. News published on non-trading days is assigned to the next trading day. The monthly macroeconomic value is carried over to every day of its month so that it aligns with the daily data. The regime labels are fitted once on the whole history and are used as descriptive inputs; refitting them in a walk-forward manner is planned as a refinement.

6.6.3 Testing and Validation Policy

Each stage is validated on its own before integration: the sentiment model on a held-out test set, the regime detector by inspecting the cluster characteristics and the silhouette score, and the forecasting model on a chronologically held-out test set against non-regime-aware baselines. The REST API and screens are tested against the use cases of Chapter 4.

6.6.4 Maintenance Policy

Scraping, scoring, forecasting and explanation are planned to run as scheduled daily jobs, and large model files are kept in cloud storage rather than in the source repository. The source code is version controlled and organised by pipeline stage, so that each component can be modified independently.

6.7 Conclusion

This chapter presented the design of FinSight. It described the layered architecture in which external data is collected, processed by a Python ML pipeline, stored in PostgreSQL and served to visitors through a REST API and a web frontend. It detailed the subsystems of the ML pipeline, the strategies that shaped the design, such as precomputing results offline, using Monte Carlo dropout for uncertainty and computing SHAP per regime, and the policies that govern data handling, testing and maintenance. The domain model was represented by the class diagram. Together these decisions give a modular, explainable and maintainable system that realises the requirements of Chapter 4 and the methodology of Chapter 5.

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